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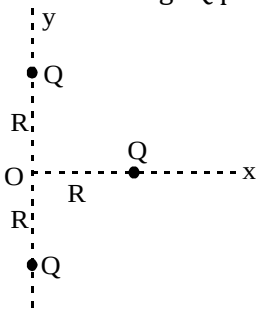
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Electrostatics

1. Find net force on charge Q placed on X-axis.

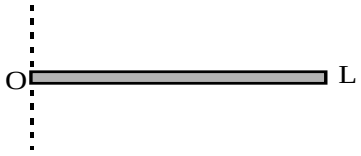


(1) $\frac{\sqrt{2}Q^2}{8\pi\epsilon_0 R^2}$ (2) $\frac{\sqrt{2}Q^2}{4\pi\epsilon_0 R^2}$

(3) $\frac{\sqrt{2}Q^2}{8\pi\epsilon_0 R}$ (4) $\frac{2\sqrt{2}Q^2}{8\pi\epsilon_0 R^2}$

PEL239

2. If linear charge density of a wire of length L depends on distance x from one end as $\lambda = \frac{\lambda_0 x}{L}$. Find total charge of wire.



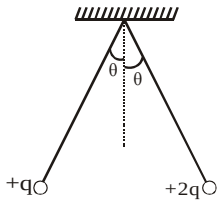
(1) $\lambda_0 L$ (2) $\frac{\lambda_0 L}{2}$

(3) $\frac{\lambda_0 L}{4}$ (4) $2\lambda_0 L$

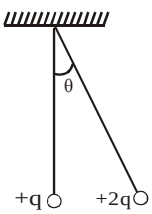
PEL240

3. Two metal spheres of same mass are suspended from a common point by a light insulating string. The length of each string is same. The spheres are given electric charges +q on one and +2q on the other. Which of the following diagram best shows the resulting positions of spheres?

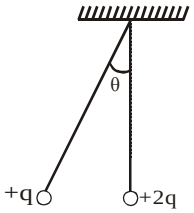
(1)



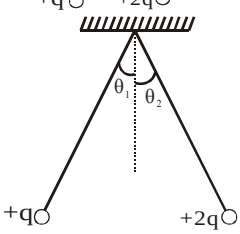
(2)



(3)



(4)



PEL241

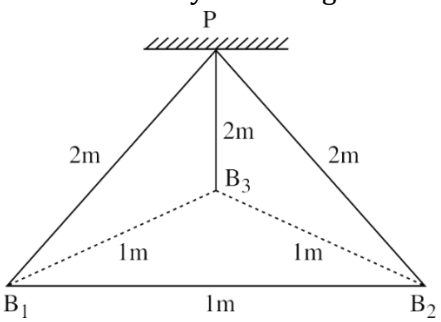
4. A vertical electric field of magnitude 4.9×10^5 N/C just prevents a water droplet of a mass 0.1 g from falling. The value of charge on the droplet will be : (Given g = 9.8 m/s²)

(1) 1.6×10^{-9} C (2) 2.0×10^{-9} C

(3) 3.2×10^{-9} C (4) 0.5×10^{-9} C

PEL242

5. Three identical charged balls each of charge 2C are suspended from a common point P by silk threads of 2m each (as shown in figure). They form an equilateral triangle of side 1m. The ratio of net force on a charged ball to the force between any two charged balls will be :



(1) 1 : 1 (2) 1 : 4

(3) $\sqrt{3}:2$ (4) $\sqrt{3}:1$

PEL243

6. Two point charges A and B of magnitude $+8\times10^{-6}$C and -8×10^{-6}C respectively are placed at a distance d apart. The electric field at the middle point O between the charges is 6.4×10^{4}NC^{-1}. The distance 'd' between the point charges A and B is:

(1) 2.0 m (2) 3.0 m

(3) 1.0 m (4) 4.0 m

PEL244

7. Two charges q and 3q are separated by a distance 'r' in air. At a distance x from charge q, the resultant electric field is zero. The value of x is :

(1) $\frac{(1+\sqrt{3})}{r}$ (2) $\frac{r}{3(1+\sqrt{3})}$

(3) $\frac{r}{(1+\sqrt{3})}$ (4) $r(1+\sqrt{3})$

PEL245

8. A positive charge particle of 100 mg is thrown in opposite direction to a uniform electric field of strength 1×10^5 NC^{-1}. If the charge on the particle is 40 μC and the initial velocity is 200 ms^{-1}, how much distance it will travel before coming to the rest momentarily :

(1) 1 m (2) 5 m

(3) 10 m (4) 0.5 m

PEL246

9. Two point charges Q each are placed at a distance d apart. A third point charge q is placed at a distance x from mid-point on the perpendicular bisector. The value of x at which charge q will experience the maximum Coulomb's force is :

(1) x = d (2) $x = \frac{d}{2}$

(3) $x = \frac{d}{\sqrt{2}}$ (4) $x = \frac{d}{2\sqrt{2}}$

PEL247

10. Three point charges of magnitude 5μC, 0.16μC and 0.3μC are located at the vertices A, B, C of a right angled triangle whose sides are AB = 3cm, $BC = 3\sqrt{2}$ cm and CA=3 cm and point A is the right angle corner. Charge at point A experiences _____ N of electrostatic force due to the other two charges.

PEL248

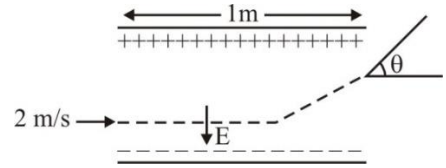
11. A charge of 4 μC is to be divided into two. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be:

(1) 1 μC and 3 μC (2) 2 μC and 2 μC

(3) 0 and 4 μC (4) 1.5 μC and 2.5 μC

PEL249

12. A uniform electric field E = (8m/e) V/m is created between two parallel plates of length 1m as shown in figure, (where m = mass of electron and e = charge of electron). An electron enters the field symmetrically between the plates with a speed of 2m/s. The angle of the deviation (θ) of the path of the electron as it comes out of the field will be .....



(1) $\tan^{-1}(4)$ (2) $\tan^{-1}(2)$

(3) $\tan^{-1}\left(\frac{1}{3}\right)$ (4) $\tan^{-1}(3)$

PEL250

13. Two identical metallic spheres A and B when placed at certain distance in air repel each other with a force of F. Another identical uncharged sphere C is first placed in contact with A and then in contact with B and finally placed at midpoint between spheres A and B. The force experienced by sphere C will be :

(1) 3F/2 (2) 3F/4 (3) F (4) 2F

PEL251

14. If two charges q_1 and q_2 are separated with distance 'd' and placed in a medium of dielectric constant K. What will be the equivalent distance between charges in air for the same electrostatic force ?

(1) $d\sqrt{K}$ (2) $K\sqrt{d}$

(3) $1.5d\sqrt{K}$ (4) $2d\sqrt{K}$

PEL252

15. A stream of a positively charged particles having $\frac{q}{m} = 2\times10^{11}\frac{C}{kg}$ and velocity $\vec{v}_0 = 3\times10^7\hat{i}\ m/s$ is deflected by an electric field $1.8\hat{j}kV/m$. The electric field exists in a region of 10 cm along x direction. Due to the electric field, the deflection of the charged particles in the y direction is ____ mm.

PEL253

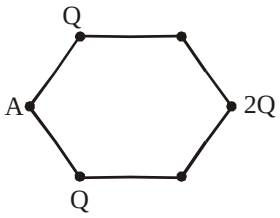
16. A rod of length L has a total charge Q distributed uniformly along its length. It is bent in the shape of a semicircle. Find the magnitude of the electric field at the centre of curvature of the semicircle.

(1) $Q/2\in_0 L^2$ (2) $Q/\in_0 L^2$

(3) $\frac{2Q}{\in_0 L^2}$ (4) Zero

PEL254

17. Regular hexagon with side length 'a' is shown in fig. Find E_A ?



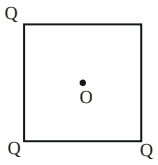
(1) $\frac{2kQ}{a^2}$ (2) $\frac{3}{2}\frac{kQ}{a^2}$

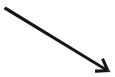
(3) $\frac{5}{4}\frac{kQ}{a^2}$ (4) $\frac{kQ}{a^2}$

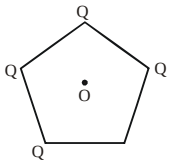
PEL255

18. In the given diagrams the direction of electric field at point O is given in list-II (O is circumcenter of the given regular polygon). Charge Q is positive. Match the direction of electric field for the given arrangement:-

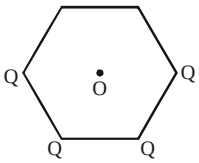
List-I **List-II**

(P) 

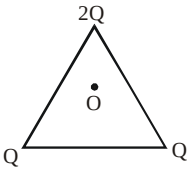
(1) 

(Q) 

(2) 

(R) 

(3) 

(S) 

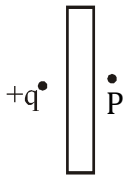
(4) 

Code :-

	P	Q	R	S
(1)	2	1	4	3
(2)	3	1	2	4
(3)	3	2	1	4
(4)	1	4	3	2

PEL256

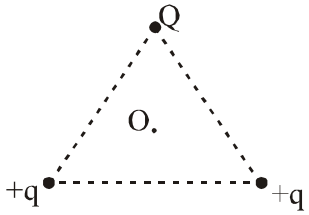
19. A point charge +q is placed in front of a neutral metal plate. The direction of electric field at point 'P' on the other side of the plate is :-



(1) Perpendicular to the plate & away from plane of plate
(2) Perpendicular to the plate & towards plane of plate
(3) Radially away from +q
(4) Radially towards +q

PEL257

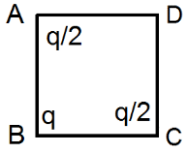
20. Three charges are placed at corners of an equilateral triangle as shown in figure. The magnitude of electric field at centre 'O' due to q is E and total field at centre O is 3E then :-



(1) Q = + 4q
(2) Q = – 2q
(3) Q = – 3q
(4) Both (1) & (2) are possible

PEL258

21. The three charges q/2, q and q/2 are placed at the corners A, B and C of a square of side 'a' as shown in figure. The magnitude of electric field (E) at the corner D of the square, is :



(1) $\frac{q}{4\pi\epsilon_0 a^2}\left(\frac{1}{\sqrt{2}}+\frac{1}{2}\right)$

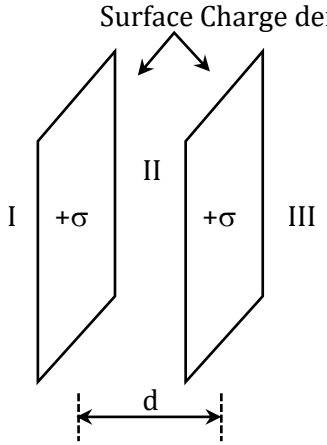
(2) $\frac{q}{4\pi\epsilon_0 a^2}\left(1+\frac{1}{\sqrt{2}}\right)$

(3) $\frac{q}{4\pi\epsilon_0 a^2}\left(1-\frac{1}{\sqrt{2}}\right)$

(4) $\frac{q}{4\pi\epsilon_0 a^2}\left(\frac{1}{\sqrt{2}}-\frac{1}{2}\right)$

PEL259

22. Let σ be the uniform surface charge density of two infinite thin plane sheets shown in figure. Then the electric fields in three different region E_I, E_{II} and E_{III} are:



(1) $\vec{E}_I = \frac{2\sigma}{\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{2\sigma}{\epsilon_0}\hat{n}$

(2) $\vec{E}_I = 0, \vec{E}_{II} = \frac{\sigma}{\epsilon_0}\hat{n}, \vec{E}_{III} = 0$

(3) $\vec{E}_I = \frac{\sigma}{2\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{2\epsilon_0}\hat{n}$

(4) $\vec{E}_I = -\frac{\sigma}{\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{\epsilon_0}\hat{n}$

PEL260

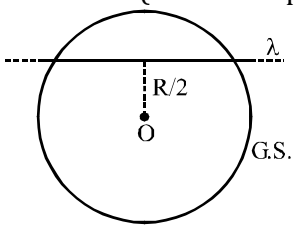
23. In a cuboid of dimension $2L \times 2L \times L$, a charge q is placed at the centre of the surface 'S' having area of 4 L^2. The flux through the opposite surface to 'S' is given by

(1) $\frac{q}{12\varepsilon_0}$ (2) $\frac{q}{3\varepsilon_0}$

(3) $\frac{q}{2\varepsilon_0}$ (4) $\frac{q}{6\varepsilon_0}$

PEL261

24. Find flux from G.S. (Radius of sphere is R)

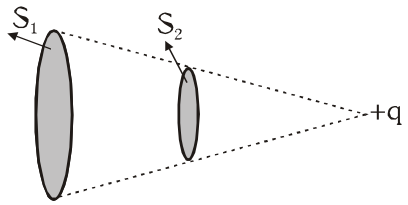


(1) $\frac{\sqrt{3}\lambda R}{2\epsilon_0}$ (2) $\frac{\sqrt{3}\lambda R}{\epsilon_0}$

(3) $\frac{\lambda R}{\epsilon_0}$ (4) $\frac{2\lambda R}{\epsilon_0}$

PEL262

25. Two surfaces S_1 & S_2 are shown in figure. Flux associated with S_1 is ϕ_1 & S_2 is ϕ_2. Which is correct?



(1) $\phi_1 > \phi_2$

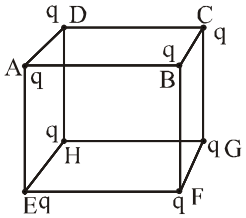
(2) $\phi_1 < \phi_2$

(3) $\phi_1 = \phi_2$

(4) can't say anything

PEL263

26. Eight point charges having magnitude q are fixed at vertices of a cube. The electric flux through square surface ABCD of the cube is

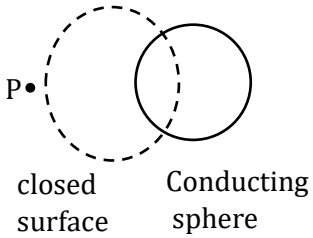


(1) $\frac{q}{24\epsilon_0}$ (2) $\frac{q}{12\epsilon_0}$

(3) $\frac{q}{6\epsilon_0}$ (4) $\frac{q}{3\epsilon_0}$

PEL264

27. Figure shows a closed surface which intersects a conducting sphere. If a positive charge is placed at the point P, the net flux of the electric field through the closed surface



(1) will remain zero

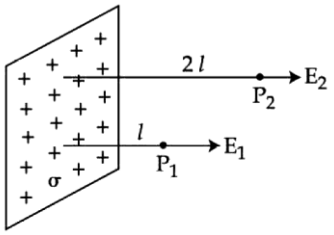
(2) will become positive

(3) will become negative

(4) will become undefined

PEL265

28. In the figure, a very large plane sheet of positive charge is shown. P_1 and P_2 are two points at distance l and $2l$ from the charge distribution. If σ is the surface charge density, then the magnitude of electric fields E_1 and E_2 at P_1 and P_2 respectively are :



(1) $E_1 = \sigma/\varepsilon_0, E_2 = \sigma/2\varepsilon_0$

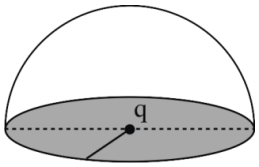
(2) $E_1 = 2\sigma/\varepsilon_0, E_2 = \sigma/\varepsilon_0$

(3) $E_1 = E_2 = \sigma/2\varepsilon_0$

(4) $E_1 = E_2 = \sigma/\varepsilon_0$

PEL266

29. If a charge q is placed at the centre of a closed hemispherical non-conducting surface, the total flux passing through the flat surface would be :



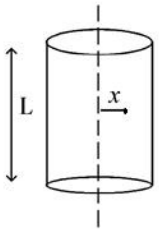
(1) $\frac{q}{\varepsilon_0}$ (2) 0 (3) $\frac{q}{4\varepsilon_0}$ (4) $\frac{q}{2\pi\varepsilon_0}$

PEL267

30. The volume charge density of a sphere of radius 6 m is 2 $\mu C\ cm^{-3}$. The number of lines of force per unit surface area coming out from the surface of the sphere is _____ $\times 10^{10}\ NC^{-1}$.
[Given : Permittivity of vacuum
$\in_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1} - m^{-2}$]

PEL268

31. A long cylindrical volume contains a uniformly distributed charge of density $\rho\ cm^{-3}$. The electric field inside the cylindrical volume at a distance $x = \frac{2\varepsilon_0}{\rho}$ m from its axis is _____ Vm^{-1}



PEL269

32. Given below are two statements.
Statement I : Electric potential is constant within and at the surface of each conductor.
Statement II : Electric field just outside a charged conductor is perpendicular to the surface of the conductor at every point.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct

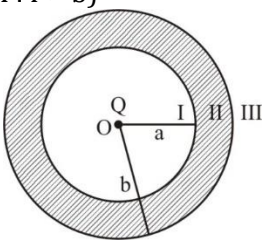
(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

PEL270

33. As shown in the figure, a point charge Q is placed at the centre of conducting spherical shell of inner radius a and outer radius b. The electric field due to charge Q in three different regions I, II and III is given by : (I : r < a, II : a < r < b, III : r > b)



(1) $E_I = 0, E_{II} = 0, E_{III} \neq 0$

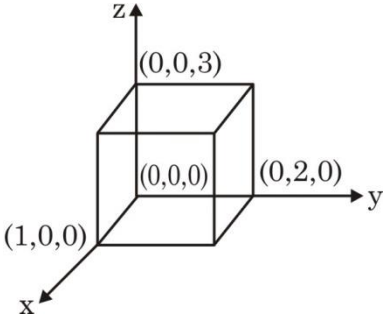
(2) $E_I \neq 0, E_{II} = 0, E_{III} \neq 0$

(3) $E_I \neq 0, E_{II} = 0, E_{III} = 0$

(4) $E_I = 0, E_{II} = 0, E_{III} = 0$

PEL271

34. As shown in figure, a cuboid lies in a region with electric field $\vec{E} = 2x^2\hat{i} - 4y\hat{j} + 6\hat{k}$ N/C. The magnitude of charge within the cuboid is $n \in_0$ C. The value of n is _____
(if dimension of cuboid is 1 × 2 × 3 m^3)

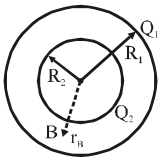


PEL272

35. Considering a group of positive charges, which of the following statements is correct?
(1) Net potential of the system cannot be zero at a point but net electric field can be zero at that point.
(2) Net potential of the system at a point can be zero but net electric field can't be zero at that point.
(3) Both the net potential and the net field can be zero at a point.
(4) Both the net potential and the net electric field cannot be zero at a point.

PEL273

36. You are given two concentric charged conducting spheres of radii R_1 and R_2 such that $R_1 > R_2$, having charges Q_1 and Q_2 respectively and uniformly distributed over its surface. Calculate E and V at point B whose distance from the centre is r_B as shown in figure :-



(1) $E = \frac{Q_1}{4\pi\in_0 r_B^2}$, $V = \frac{Q_1}{4\pi\in_0 R_1}$

(2) $E = \frac{Q_2}{4\pi\in_0 r_B^2}$, $V = \frac{Q_1}{4\pi\in_0 R_1} + \frac{Q_2}{4\pi\in_0 r_B}$

(3) $E = \frac{(Q_1+Q_2)}{4\pi\in_0 r_B^2}$, $V = \frac{Q_1}{4\pi\in_0 R_1} + \frac{Q_2}{4\pi\in_0 r_B}$

(4) E = 0, V = 0

PEL274

37. Eight equal charges q are placed at each corner of a cube of side a each. Work done in carrying a charge –q from its centre to infinity is

(1) zero (2) $\frac{3\sqrt{2}q^2}{\pi\in_0 A}$

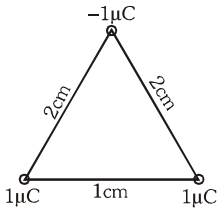
(3) $\frac{\sqrt{2}q^2}{\pi\in_0 a}$ (4) $\frac{4q^2}{\sqrt{3}\pi\in_0 a}$

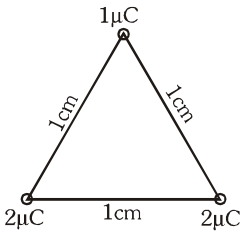
PEL275

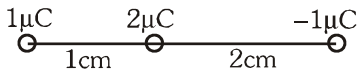
38. The electric field intensity at a point at a distance 2 m from a charge q is 4N/C. Find the amount of work done (in joule) in bringing a charge of 1 coulomb from infinity to this point.
(1) 2 J (2) 4 J (3) 8 J (4) 16 J

PEL276

39. Which of the following systems of charges have zero electrostatic potential energy?

(1)


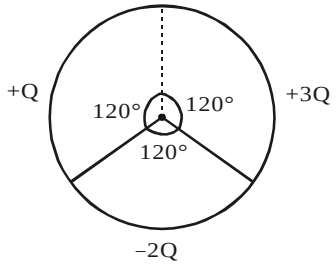
(2)


(3)


(4) None of these

PEL277

40. Figure shows three circular arcs, each of radius R and total charge as indicated. The net electric potential at the centre of curvature is



(1) $\frac{Q}{4\pi\in_0 R}$ (2) $\frac{Q}{2\pi\in_0 R}$

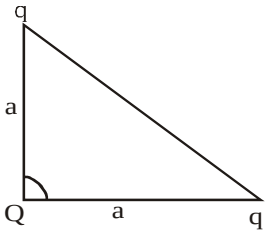
(3) $\frac{2Q}{\pi\in_0 R}$ (4) $\frac{Q}{\pi\in_0 R}$

PEL278

41. A hollow metal sphere of radius 5 cm is charged such that the potential on its surface is 10 V. The potential at the centre of the sphere is
(1) 0 V
(2) 10 V
(3) same as at point 5 cm away from the surface outside sphere.
(4) same as a point 25 cm away from the surface.

PEL279

42. Three charges Q, q & q are placed at the vertices of a right angle isosceles triangle as shown. The net electrostatic energy of the configuration is zero, if q/Q is equal to



(1) –1 (2) $-\sqrt{2}$

(3) –2 (4) $-2\sqrt{2}$

PEL280

43. Three point charge q, –2q and –2q are placed at the vertices of an equilateral triangle of side a. The minimum work done by some external force to increase their separation to 2a will be

(1) $\frac{1}{4\pi\in_0}\frac{2q^2}{a}$ (2) negative

(3) zero (4) $\frac{1}{4\pi\in_0}\frac{3q^2}{a}$

PEL281

44. A solid sphere of radius R is charged uniformly. At what distance from its surface is the electrostatic potential half of the potential at the centre?

(1) R (2) R/2 (3) R/3 (4) 2R

PEL282

45. Three equal charges Q are placed at the vertices of an equilateral triangle. What should be the value of a charge so that when placed at the centroid, self energy of the system becomes zero.

(1) $\frac{-Q}{2}$ (2) $\frac{-Q}{3}$ (3) $\frac{-Q}{2\sqrt{3}}$ (4) $\frac{-Q}{\sqrt{3}}$

PEL283

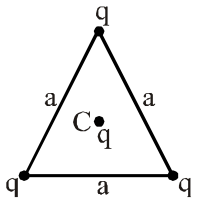
46. The force acting between two point charges + q and + q separated by distance r is F. If one charge is fixed and another charge completes two rounds in circular path around it, then work done will be:-

(1) F × 4 π r (2) F × 2r

(3) Zero (4) F × πr^2

PEL284

47. Four equal charges of magnitude q are placed as shown in figure. Now the charge at center C is taken to infinity slowly, then work done by external force will be -



(1) $\frac{-3kq^2}{a}$ (2) $\frac{-3\sqrt{3}kq^2}{a}$

(3) $\frac{-3kq^2}{a}(\sqrt{3}+1)$ (4) $\frac{3kq^2}{a}(\sqrt{3}+1)$

PEL285

48. If the electric potential at any point (x, y, z)m in space is given by V = $3x^2$ volt. The electric field at the point (1, 0, 3) m will be :

(1) 3 Vm^{-1}, directed along positive x-axis.

(2) 3 Vm^{-1}, directed along negative x-axis.

(3) 6 Vm^{-1}, directed along positive x-axis.

(4) 6 Vm^{-1}, directed along negative x-axis.

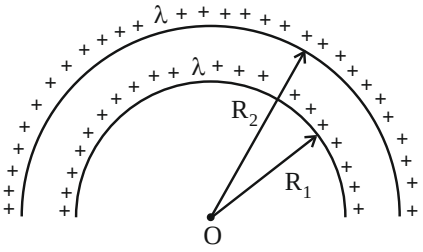
PEL286

49. Two uniformly charged spherical conductors A and B of radii 5 mm and 10 mm are separated by a distance of 2 cm. If the spheres are connected by a conducting wire, then in equilibrium condition, the ratio of the magnitudes of the electric fields at the surface of the sphere A and B will be :

(1) 1 : 2 (2) 2 : 1 (3) 1 : 1 (4) 1 : 4

PEL287

50. The electric potential at the centre of two concentric half rings of radii R_1 and R_2, having same linear charge density λ is



(1) $\frac{2\lambda}{\in_0}$ (2) $\frac{\lambda}{2\in_0}$ (3) $\frac{\lambda}{4\in_0}$ (4) $\frac{\lambda}{\in_0}$

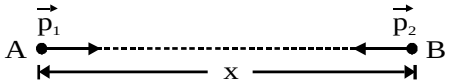
PEL288

51. Two isolated metallic solid spheres of radii R and 2R are charged such that both have same charge density σ. The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ'. The ratio $\frac{\sigma'}{\sigma}$ is :

(1) $\frac{9}{4}$ (2) $\frac{4}{3}$ (3) $\frac{5}{3}$ (4) $\frac{5}{6}$

PEL289

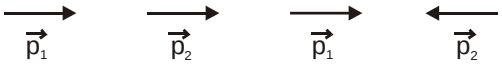
52. Two small electric dipoles, one of dipole moment $\vec{p}_1$ at point A and the other of dipole moment $\vec{p}_2$ at point B, are as shown in the figure. The torque experienced by the dipole $\vec{p}_2$ is :-



(1) Zero (2) ∞
(3) Can't be determined (4) None of these

PEL290

53. For shown situation of two dipoles the nature of forces between them are :-



(I) **(II)**

(1) attraction, attraction
(2) attraction, repulsion
(3) repulsion, repulsion
(4) repulsion, attraction

PEL291

54. The force of attraction between two co-axial dielectric dipoles whose centers are r meters apart varies with distance as

(1) r^{-1} (2) r^{-2} (3) r^{-3} (4) r^{-4}

PEL292

55. A and B are two points on the axis and the perpendicular bisector respectively of an electric dipole. A and B are far away from the dipole and at equal distance from it. The field at A and B are $\vec{E}_A$ and $\vec{E}_B$.

(1) $\vec{E}_A = \vec{E}_B$

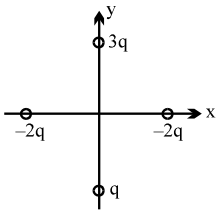
(2) $\vec{E}_A = 2\vec{E}_B$

(3) $\vec{E}_A = -2\vec{E}_B$

(4) $E_B = \frac{1}{2}|E_A$,and $\vec{E}_B$ is perpendicular to $\vec{E}_A$

PEL293

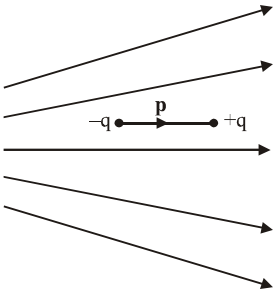
56. 4 charges are placed each at a distance 'a' from origin. The dipole moment of configuration is



(1) $2qa\hat{j}$ (2) $3qa\hat{j}$
(3) $2aq[\hat{i}+\hat{j}]$ (4) none

PEL294

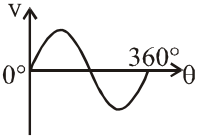
57. Figure shows electric field lines in which an electric dipole **p** is placed as shown. Which of the following statements is correct?

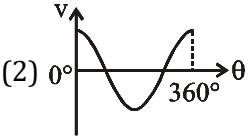


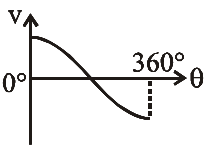
(1) The dipole will not experience any force.
(2) The dipole will experience a force towards right.
(3) The dipole will experience a force towards left.
(4) The dipole will experience a force upwards.

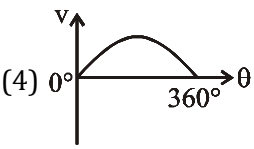
PEL295

58. Consider a point A far away from short dipole p. By keeping r constant, θ varies from 0° to 360°. Variation of potential of dipole at A with respect to θ :-

(1) 

(2) 

(3) 

(4) 

PEL296

59. Electric field at a far away distance r on the axis of dipole is $\vec{E}_0$. What is the electric field at a distance 2r on perpendicular bisector?

(1) $\frac{\vec{E}_0}{16}$ (2) $-\frac{\vec{E}_0}{16}$ (3) $\frac{\vec{E}_0}{8}$ (4) $-\frac{\vec{E}_0}{8}$

PEL297

60. Given below are two statements :

Statement-I : A point charge is brought in an electric field. The value of electric field at a point near to the charge may increase if the charge is positive.

Statement-II : An electric dipole is placed in a non-uniform electric field. The net electric force on the dipole will not be zero.

Choose the correct answer from the options given below :

(1) Both statement-I and statement-II are true.

(2) Both statement-I and statement-II are false.

(3) Statement-I is true but statement-II is false.

(4) Statement-I is false but statement-II is true.

PEL298

61. Two electric dipoles of dipole moments 1.2×10^{-30} cm and 2.4×10^{-30} cm are placed in two different uniform electric fields of strengths 5×10^4 NC^{-1} and 15×10^4 NC^{-1} respectively. The ratio of maximum torque experienced by the electric dipoles will be $\frac{1}{x}$.

The value of x is _____.

PEL299

62. In a certain region of space, the potential is given by : $V = k[2x^2 - y^2 + z^2]$. The electric field at the point (1, 1, 1) has magnitude =

(1) $k\sqrt{6}$ (2) $2k\sqrt{6}$

(3) $2k\sqrt{3}$ (4) $4k\sqrt{3}$

PEL300

63. Electric field in a region is directed outwards and proportional to distance r from origin. Taking potential at origin to be zero, potential in the region is :

(1) Uniform

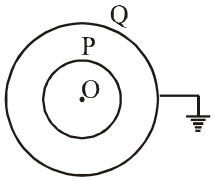
(2) Proportional to r

(3) Proportional to r^2

(4) Increases as we move away from origin.

PEL301

64. P and Q are two concentric metallic shells. If P is positively charged and Q is earthed. then:-



(1) charge density on Q is same as on P

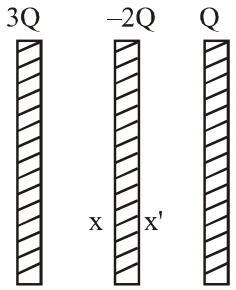
(2) electric field between P & Q is uniform

(3) electric potential inside P is zero

(4) electric field outside Q is zero

PEL302

65. Charge on both surfaces of middle metal plate:



(1) x = – 2Q, x' = 0

(2) x = + 2Q, x' = 0

(3) x = – 2Q, x' = + 4 Q

(4) x = – 4Q, x' = + 2Q

PEL303

66. 64 small drops of mercury, each of radius r and charge q are mixed together to form a big drop. The ratio of surface charge density of each small drop to big drop is :-

(1) 64 : 1

(2) 1 : 64

(3) 1 : 4

(4) 4 : 1

PEL304

ANSWER KEY

1.	(1)	**2.**	(2)	**3.**	(1)	**4.**	(2)	**5.**	(4)
6.	(2)	**7.**	(3)	**8.**	(4)	**9.**	(4)	**10.**	17
11.	(2)	**12.**	(2)	**13.**	(2)	**14.**	(1)	**15.**	2
16.	(1)	**17.**	(2)	**18.**	(2)	**19.**	(1)	**20.**	(4)
21.	(1)	**22.**	(4)	**23.**	(4)	**24.**	(2)	**25.**	(3)
26.	(3)	**27.**	(3)	**28.**	(3)	**29.**	(2)	**30.**	45
31.	1	**32.**	(1)	**33.**	(2)	**34.**	12	**35.**	(1)
36.	(2)	**37.**	(4)	**38.**	(3)	**39.**	(1)	**40.**	(2)
41.	(2)	**42.**	(4)	**43.**	(3)	**44.**	(3)	**45.**	(4)
46.	(3)	**47.**	(2)	**48.**	(4)	**49.**	(2)	**50.**	(2)
51.	(4)	**52.**	(1)	**53.**	(2)	**54.**	(4)	**55.**	(3)
56.	(1)	**57.**	(3)	**58.**	(2)	**59.**	(2)	**60.**	(1)
61.	6	**62.**	(2)	**63.**	(3)	**64.**	(4)	**65.**	(1)
66.	(3)								

Current Electricity

1. A potential difference V exists between the ends of a metal wire of length ℓ. The drift velocity will be doubled if :-

(1) V is doubled

(2) ℓ is doubled

(3) The diameter of the wire is doubled

(4) The temperature of the wire is doubled

PCE268

2. Three copper wires have their lengths in the ratio 5 : 3 : 1 and their masses are in the ratio 1 : 3 : 5. Their electrical resistance will be in the ratio

(1) 5 : 3 : 1 (2) 1 : 3 : 5

(3) 125 : 15 : 1 (4) 1 : 15 : 125

PCE269

3. A cylindrical copper wire is stretched such that its diameter decreases by 0.01%. The approximate percentage increase in its resistance is :-

(1) 0.02% (2) 0.04%

(3) 0.08% (4) 0.16%

PCE270

4. The resistance of wire is 5Ω at 50°C & 6Ω at 100°C. The resistance of the wire at 0°C will be :-

(1) 2Ω (2) 1Ω (3) 4Ω (4) 3Ω

PCE271

5. The drift velocity of electrons for a conductor connected in an electrical circuit is V_d. The conductor in now replaced by another conductor with same material and same length but double the area of cross section. The applied voltage remains same. The new drift velocity of electrons will be

(1) V_d (2) $\frac{V_d}{2}$

(3) $\frac{V_d}{4}$ (4) $2V_d$

PCE272

6. The length of a given cylindrical wire is increased to double of its original length. The percentage increase in the resistance of the wire will be ___%.

PCE273

7. The current density in a cylindrical wire of radius 4 mm is $4\times10^6\,Am^{-2}$. The current through the outer portion of the wire between radial distance $\frac{R}{2}$ and R is ___ π A.

PCE274

8. Resistance of the wire is measured as 2Ω and 3Ω at 10°C and 30°C respectively. Temperature co–coefficient of resistance of the material of the wire is :

(1) $0.033°C^{-1}$ (2) $-0.033°C^{-1}$

(3) $0.011°C^{-1}$ (4) $0.055°C^{-1}$

PCE275

9. A. The drift velocity of electrons decreases with the increase in the temperature of conductor.

B. The drift velocity is inversely proportional to the area of cross-section of given conductor.

C. The drift velocity does not depend on the applied potential difference to the conductor.

D. The drift velocity of electron is inversely proportional to the length of the conductor.

E. The drift velocity increases with the increase in the temperature of conductor.

Choose the correct answer from the options given below:

(1) A and B (2) A, B and D

(3) B and E (4) B and C

PCE276

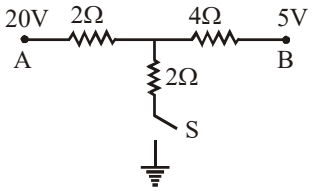
10. A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is:

(1) 9 : 1 (2) 1 : 9

(3) 4 : 1 (4) 3 : 1

PCE277

11. As the switch S is closed in the circuit shown in figure, current passed through it is :-

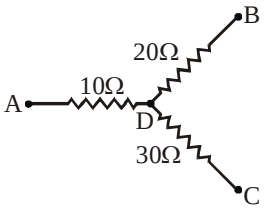


(1) 4.5 A (2) 6.0 A

(3) 3.0 A (4) Zero

PCE278

12. In the circuit given here, the points A, B and C are 70 V, 0, 10V respectively. Then



(1) the point D will be at a potential of 60V

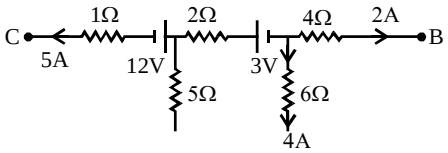
(2) the point D will be at a potential of 20V

(3) currents in the paths AD, DB and DC are in the ratio of 9:2:3

(4) currents in the paths AD, DB and DC are in the ratio of 3:2:1

PCE279

13. The figure shows part of certain circuit, find :



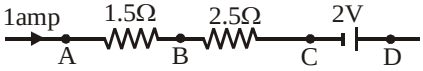
Potential difference $V_C - V_B$.

(1) 6 V (2) 10 V

(3) 4 V (4) 8 V

PCE280

14. In the circuit element given here, if the potential at point B, $V_B = 0$, then the potentials of A and D are given as:-



(1) $V_A = -1.5V$, $V_D = +2V$

(2) $V_A = +1.5V$, $V_D = +2V$

(3) $V_A = +1.5V$, $V_D = +0.5V$

(4) $V_A = +1.5V$, $V_D = -0.5V$

PCE281

15. What will be the most suitable combination of three resistors A = 2Ω, B = 4Ω, C = 6 Ω so that $\left(\frac{22}{3}\right)\Omega$ is equivalent resistance of combination?

(1) Parallel combination of A and C connected in series with B.

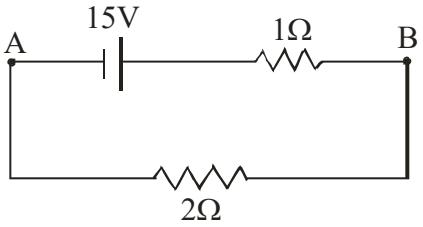
(2) Parallel combination of A and B connected in series with C.

(3) Series combination of A and C connected in parallel with B.

(4) Series combination of B and C connected in parallel with A.

PCE282

16. For the network shown below, the value $V_B - V_A$ is _____V.



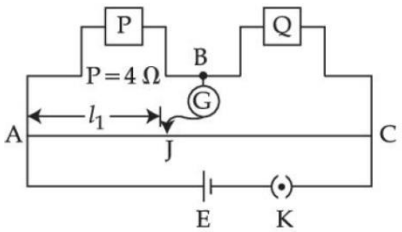
PCE283

17. The combination of two identical cells, whether connected in series or parallel combination provides the same current through an external resistance of 2Ω. The value of internal resistance of each cell is :

(1) 2Ω (2) 4Ω (3) 6Ω (4) 8Ω

PCE284

18. Resistance are connected in a meter bridge circuit as shown in the figure. The balancing length l_1 is 40cm. Now an unknown resistance x is connected in series with P and new balancing length is found to be 80cm measured from the same end. Then the value of x will be ____Ω



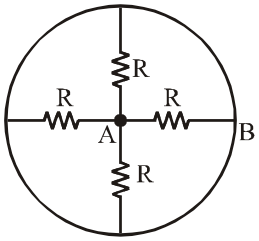
PCE285

19. Two sources of equal emfs are connected in series. This combination is connected to an external resistance R. The internal resistances of the two sources are r_1 and r_2 ($r_1 > r_2$). If the potential difference across the source of internal resistance r_1 is zero then the value of R will be

(1) $r_1 - r_2$ (2) $\frac{r_1 r_2}{r_1 + r_2}$

(3) $\frac{r_1 + r_2}{2}$ (4) $r_2 - r_1$

PCE286

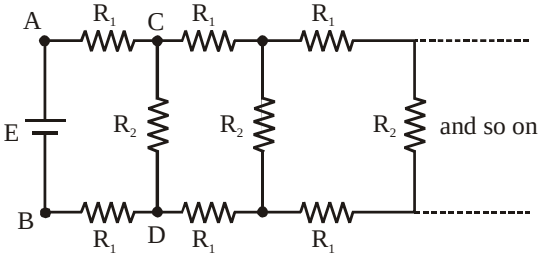
20. 

R_{AB} = ?

(1) R/4 (2) R/2 (3) R (4) 2R

PCE287

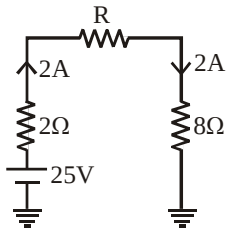
21. Find the equivalent resistance between A and B of the network extending off to the infinity shown in the figure for $R_1 = 1\Omega$, $R_2 = 4\Omega$:-



(1) 4 Ω (2) 1 Ω (3) 2 Ω (4) 8 Ω

PCE288

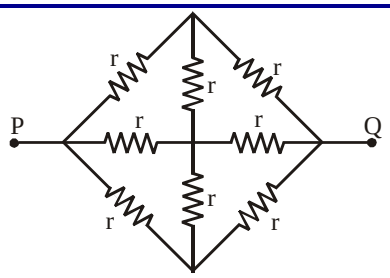
22. The value of R in the circuit should be :-



(1) 1 Ω (2) 1.5 Ω (3) 2 Ω (4) 2.5 Ω

PCE289

23. The equivalent resistance between the points P and Q in the network given here is equal to (given $r = \frac{3}{2}\Omega$):-



(1) $\frac{1}{2}\Omega$ (2) 1Ω

(3) $\frac{3}{2}\Omega$ (4) 2Ω

PCE290

24. n identical cells are joined in series with its two cells A and B in the loop with reversed polarities. EMF of each cell is E and internal resistance r. Potential difference across cell A or B is (here n > 4)

(1) $\frac{2E}{n}$ (2) $2E\left(1-\frac{1}{n}\right)$

(3) $\frac{4E}{n}$ (4) $2E\left(1-\frac{2}{n}\right)$

PCE291

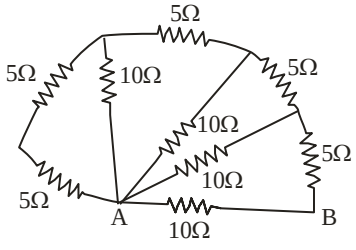
25. Two cells of equal e.m.f. and of internal resistances r_1 and $r_2(r_1 > r_2)$ are connected in series. On connecting this combination to an external resistance R, it is observed that the potential difference across the first cell becomes zero. The value of R will be

(1) $r_1 + r_2$ (2) $r_1 - r_2$

(3) $\frac{r_1 + r_2}{2}$ (4) $\frac{r_1 - r_2}{2}$

PCE292

26. The equivalent resistance between points A and B in the given network is :

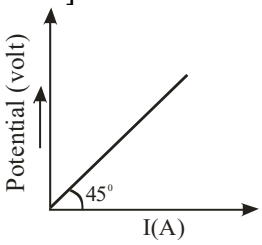


(1) 65Ω (2) 20Ω

(3) 5Ω (4) 2Ω

PCE293

27. The variation of applied potential and current flowing through a given wire is shown in figure. The length of wire is 31.4 cm. The diameter of wire is measured as 2.4 cm. The resistivity of the given wire is measured as $x \times 10^{-3}$ Ω cm. The value of x is _____.
[Take π = 3.14]



PCE294

28. A battery of internal resistance 2Ω is connected to a variable resistor whose value can vary from 4Ω to 10 Ω. The resistance is initially set at 4Ω. If the resistance is now increased then–
(1) power consumed by it will decrease
(2) power consumed by it will increase
(3) power consumed by it may increase or may decrease
(4) power consumed will first increase then decrease

PCE295

29. n identical bulbs are rated P watt and V volt. When they are connected in series, total power consumed is 20 watt and when connected in parallel they consume total power of 2000 watt. How many bulbs are used :-
(1) 200 bulbs (2) 100 bulbs
(3) 20 bulbs (4) 10 bulbs

PCE296

30. An electric tea kettle has two heating coils. When one of the two coils is switched on, the tea begins to boil in 10 minutes. When second coil only is switched on, the boiling begins in 15 minutes. When both coils are connected in parallel and switched on the tea starts boiling in time t, then t is :-
(1) 4 minute (2) 6 minute
(3) 25 minute (4) 12 minute

PCE297

31. Two electric bulbs whose resistance are in the ratio of 1 : 2, are connected in series to a constant voltage source. The power dissipated in them has the ratio :-
(1) 1 : 2 (2) 1 : 1
(3) 1 : 4 (4) 2 : 1

PCE298

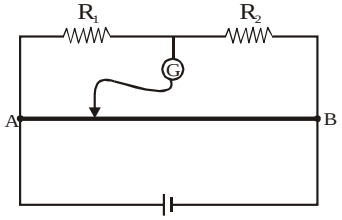
32. If two bulbs of wattage 60 W and 100 W respectively each rated at 110 V are connected in series with the supply of 220 V, which bulb will fuse ?
(1) 60 W bulb (2) 100 W bulb
(3) Both the bulbs (4) Bulbs will not fuse

PCE299

33. A resistor develops 300 J of thermal energy in 15s, when a current of 2A is passed through it. If the current increases to 3A, the energy developed in 10s is _____ J.

PCE300

34. In the figure shown for gives values of R_1 and R_2 the balance point for Jockey is at 40 cm from A. When R_2 is shunted by a resistance of 10Ω, balanced shifts to 50 cm. R_1 and R_2 are (AB = 1m):



(1) $\frac{10}{3}$ Ω, 5Ω (2) 20Ω, 30Ω
(3) 10Ω, 15Ω (4) 5Ω, $\frac{15}{2}$ Ω

PCE301

ANSWER KEY

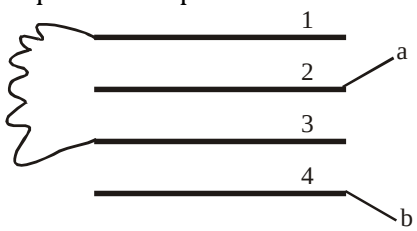
1.	(1)	**2.**	(3)	**3.**	(2)	**4.**	(3)	**5.**	(1)
6.	300	**7.**	48	**8.**	(1)	**9.**	(2)	**10.**	(1)
11.	(1)	**12.**	(4)	**13.**	(1)	**14.**	(4)	**15.**	(2)
16.	10	**17.**	(1)	**18.**	20	**19.**	(1)	**20.**	(1)
21.	(1)	**22.**	(4)	**23.**	(2)	**24.**	(4)	**25.**	(2)
26.	(3)	**27.**	144	**28.**	(1)	**29.**	(4)	**30.**	(2)
31.	(1)	**32.**	(1)	**33.**	450	**34.**	(1)		

Capacitor

1. If the plates of a parallel plate capacitor are not equal in area, then quantity of charge :-
(1) On the plates will be same but nature of charge will differ
(2) On the plates as well as nature of charge will be different
(3) On the plates will be different but nature of charge will be same
(4) As well as nature of charge will be same

PCP168

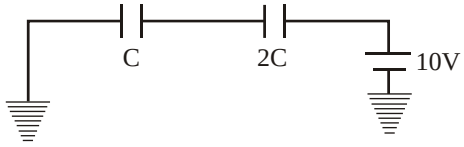
2. Consider the following arrangement of four plates interconnected as follows. Each plate has the area A and the plate separation is d. Here plates 1 and 3 are interconnected. Find the equivalent capacitance between a and b :



(1) $\frac{3}{2}\frac{\epsilon_0 A}{d}$ (2) $\frac{2}{3}\frac{\epsilon_0 A}{d}$
(3) $\frac{3\epsilon_0 A}{d}$ (4) $\frac{1}{3}\frac{\epsilon_0 A}{d}$

PCP169

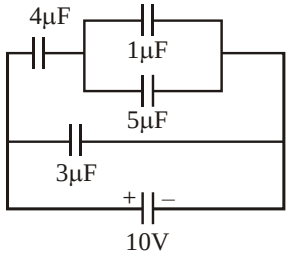
3. In the circuit shown in the figure, C = 6 μF. The charge stored in the capacitor of capacity C is:



(1) zero (2) 90 μC
(3) 40 μC (4) 60 μC

PCP170

4. For the circuit shown in the figure, the charge on 4μF capacitor is :-



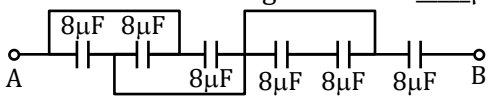
(1) 30μC (2) 40μC
(3) 24μC (4) 54μC

PCP171

5. If the charge on a capacitor is increased by 2 C, the energy stored in it increases by 44%. The original charge on the capacitor is (in C) :
(1) 10 (2) 20 (3) 30 (4) 40

PCP172

6. The equivalent capacitance between points A and B in below shown figure will be ____μF.

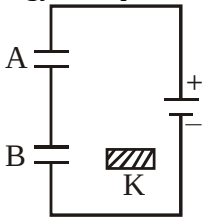


PCP173

7. A force of 10N acts on a charged particle placed between two plates of a charged capacitor. If one plate of capacitor is removed, then the force acting on that particle will be :
(1) 5 N (2) 10 N
(3) 20 N (4) Zero

PCP174

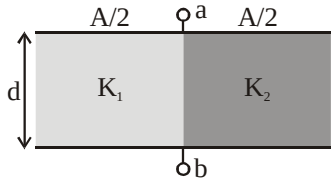
8. Two identical capacitors A and B shown in the given circuit are joined in series with a battery. If a dielectric slab of dielectric constant K is slipped between the plates of capacitors B and battery remains connected, then the energy of capacitor A will :-



(1) decrease
(2) increase
(3) remain the same
(4) be zero since circuit will not work

PCP175

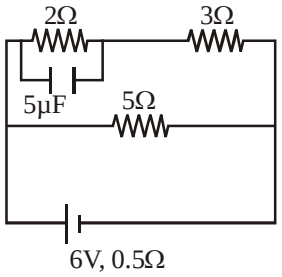
9. The capacity of a parallel plate air capacitor is 10μF. As shown in the figure this capacitor is divided into two equal parts, these parts are filled by media of dielectric constants K_1=2 and K_2=4, capacity of this arrangement will be:



(1) 20 μF (2) 30 μF
(3) 10 μF (4) 40 μF

PCP176

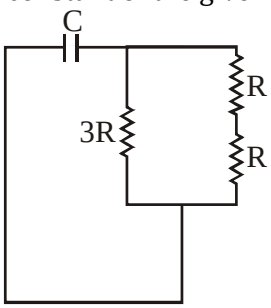
10. Find out amount of charge on capacitor shown in circuit.



(1) 0μC (2) 30μC
(3) 10μC (4) 25μC

PCP177

11. The time constant of the given circuit is :



(1) $\frac{3RC}{5}$

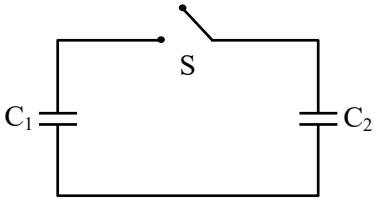
(2) $\frac{6RC}{5}$

(3) $\frac{5RC}{6}$

(4) None of the above

PCP178

12. Two capacitors having capacitance C_1 and C_2 respectively are connected as shown in figure. Initially, capacitor C_1 is charged to a potential difference V volt by a battery. The battery is then removed and the charged capacitor C_1 is now connected to uncharged capacitor C_2 by closing the switch S. The amount of charge on the capacitor C_2, after equilibrium is :



(1) $\frac{C_1C_2}{(C_1+C_2)}V$ (2) $\frac{(C_1+C_2)}{C_1C_2}V$

(3) $(C_1 + C_2)V$ (4) $(C_1 - C_2)V$

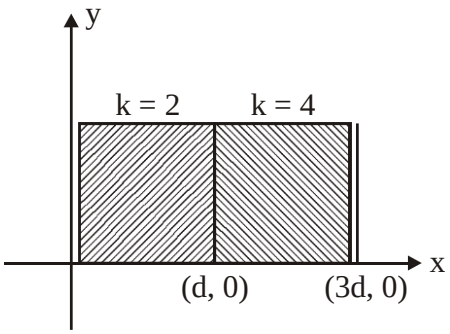
PCP179

13. Two capacitors, each having capacitance 40 μF are connected in series. The space between one of the capacitors is filled with dielectric material of dielectric constant K such that the equivalent capacitance of the system became 24 μF. The value of K will be:

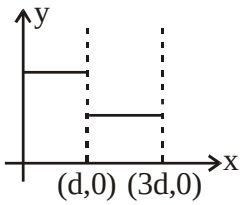
(1) 1.5 (2) 2.5
(3) 1.2 (4) 3

PCP180

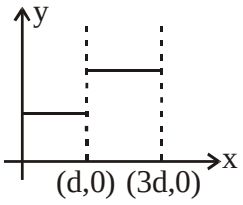
14. A parallel plate capacitor has two layers of dielectric as shown in figure. This capacitor is connected across a battery. The graph which shows the variation of electric field (E) and distance (x) from left plate :-



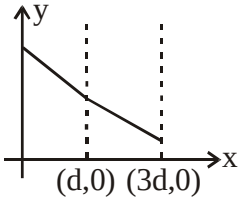
(1)



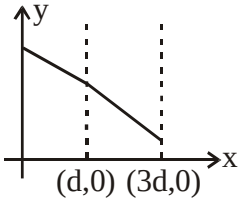
(2)



(3)



(4)



PCP181

15. A parallel plate capacitor filled with a medium of dielectric constant 10, is connected across a battery and is charged. The dielectric slab is replaced by another slab of dielectric constant 15. Then the energy of capacitor will :

(1) increase by 50%

(2) decrease by 15%

(3) increase by 25%

(4) increase by 33%

PCP182

ANSWER KEY

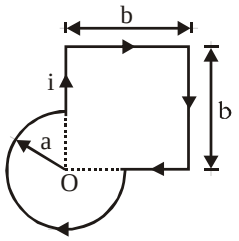
1.	(1)	**2.**	(2)	**3.**	(3)	**4.**	(3)	**5.**	(1)
6.	6	**7.**	(1)	**8.**	(2)	**9.**	(2)	**10.**	(3)
11.	(2)	**12.**	(1)	**13.**	(1)	**14.**	(1)	**15.**	(1)

Magnetic Effect of Current and Magnetism-1

1. The magnetic induction at a point P which is distant 4 cm from a long current carrying wire is 10^{-8} Tesla. Magnetic induction at a distance 12 cm from the same current carrying wire would be
(1) 3.33×10^{-9} Tesla
(2) 1.11×10^{-4} Tesla
(3) 3×10^{-3} Tesla
(4) 9×10^{-2} Tesla

PME302

2. The magnetic field at the point O of a loop with current i, whose shape is illustrated below is :-



(1) $\frac{\mu_0 i}{4\pi}\left[\frac{3\pi}{2a}+\frac{\sqrt{2}}{b}\right]$ (2) $\frac{\mu_0 i}{4\pi^2}\left[\frac{2}{b}+b\right]$
(3) $\frac{\mu_0 i}{2\pi^2}\left[\frac{1}{a}+\frac{1}{b}\right]$ (4) $\frac{\mu_0 i}{4\pi}\left[\frac{1}{a}+\frac{1}{b}\right]$

PME303

3. Two concentric circular coils of ten turns each are situated in the same plane. Their radii are 20 and 40 cm and they carry respectively 0.2 and 0.3 ampere current in opposite direction. The magnetic field in weber/m^2 at the centre is

(1) $\frac{35}{4}\mu_0$ (2) $\frac{\mu_0}{80}$
(3) $\frac{7}{80}\mu_0$ (4) $\frac{5}{4}\mu_0$

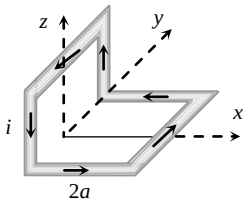
PME304

4. A cell is connected between two points of a uniformly thick circular conductor. The magnetic field at the centre of the loop will be (Here i_1 and i_2 are the currents flowing in the two parts of the circular conductor of radius 'a' and μ_0 has the usual meaning)

(1) Zero (2) $\frac{\mu_0}{2a}(i_1-i_2)$
(3) $\frac{\mu_0}{2a}(i_1+i_2)$ (4) $\frac{\mu_0}{a}(i_1+i_2)$

PME305

5. A non-planar loop of conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length 2a. The magnetic field due to this loop at the point P (a,0,a) points in the direction.



(1) $\frac{1}{\sqrt{2}}\left(-\hat{j}+\hat{k}\right)$ (2) $\frac{1}{\sqrt{3}}\left(-\hat{j}+\hat{k}+\hat{i}\right)$
(3) $\frac{1}{\sqrt{3}}\left(\hat{i}+\hat{j}+\hat{k}\right)$ (4) $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{k}\right)$

PME306

6. Two long current carrying conductors are placed parallel to each other at a distance of 8 cm between them. The magnitude of magnetic field produced at mid-point between the two conductors due to current flowing in them is 300 μT. The equal current flowing in the two conductors is :
(1) 30A in the same direction.
(2) 30A in the opposite direction.
(3) 60A in the opposite direction.
(4) 300A in the opposite direction.

PME307

7. Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is 6 : 5 and their respective masses ratio is 9 : 4. Then, the ratio of their charges will be:
(1) 8 : 5 (2) 5 : 4 (3) 5 : 3 (4) 8 : 7

PME308

8. The electric current in a circular coil of 2 turns produces a magnetic induction B_1 at its centre. The coil is unwound and is rewound into a circular coil of 5 turns and the same current produces a magnetic induction B_2 at its centre. The ratio of $\frac{B_2}{B_1}$ is :

(1) $\frac{5}{2}$ (2) $\frac{25}{4}$ (3) $\frac{5}{4}$ (4) $\frac{25}{2}$

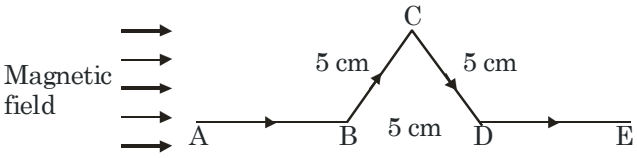
PME309

9. A charge particle is moving in a uniform magnetic field $\left(2\hat{i}+3\hat{j}\right)T$. If it has an acceleration of $\left(\alpha\hat{i}-4\hat{j}\right)m/s^2$, then the value of α will be

(1) 3 (2) 6 (3) 12 (4) 2

PME310

10. A triangular shaped wire carrying 10A current is placed in a uniform magnetic field of 0.5T, as shown in figure. The magnetic force on segment CD is (Given BC = CD = BD = 5 cm).



(1) 0.126 N (2) 0.312 N

(3) 0.216 N (4) 0.245 N

PME311

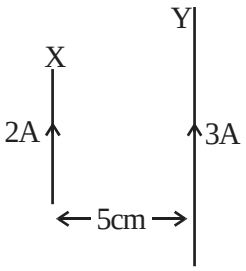
11. The magnetic field at the center of current carrying circular loop is B_1. The magnetic field at a distance of $\sqrt{3}$ times radius of the given circular loop from the center on its axis is B_2. The value of B_1/B_2 will be

(1) 9 : 4 (2) $12:\sqrt{5}$

(3) 8 : 1 (4) $5:\sqrt{3}$

PME312

12. A wire X of length 50 cm carrying a current of 2 A is placed parallel to a long wire Y of length 5 m. The wire Y carries a current of 3 A. The distance between two wires is 5 cm and currents flow in the same direction. The force acting on the wire Y is :



(1) 1.2×10^{-5} N directed towards wire X.

(2) 1.2×10^{-4} N directed away from wire X.

(3) 1.2×10^{-4} N directed towards wire X.

(4) 2.4×10^{-5} N directed towards wire X.

PME313

13. A charge particle of 2 μC accelerated by a potential difference of 100V enters a region of uniform magnetic field of magnitude 4 mT at right angle to the direction of field. The charge particle completes semicircle of radius 3 cm inside magnetic field. The mass of the charge particle is____ $\times 10^{-18}$ kg.

PME314

14. Two long straight wires P and Q carrying equal current 10A each were kept parallel to each other at 5 cm distance. Magnitude of magnetic force experienced by 10 cm length of wire P is F_1. If distance between wires is halved and currents on them are doubled, force F_2 on 10 cm length of wire P will be :

(1) 8 F_1 (2) 10 F_1

(3) $F_1/8$ (4) $F_1/10$

PME315

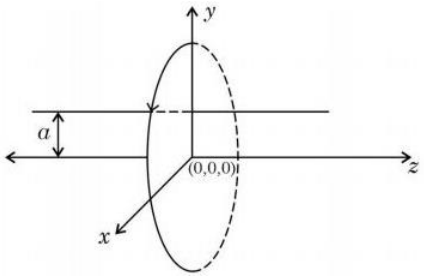
15. A circular loop of radius r is carrying current 1A. The ratio of magnetic field at the centre of circular loop and at a distance r from the center of the loop on its axis is :

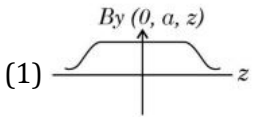
(1) $1:3\sqrt{2}$ (2) $3\sqrt{2}:2$

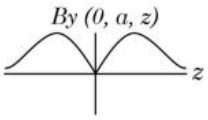
(3) $2\sqrt{2}:1$ (4) $1:\sqrt{2}$

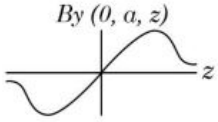
PME316

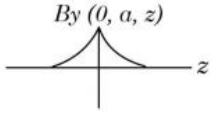
16. A single current carrying loop of wire carrying current I flowing in anticlockwise direction seen from +ve z direction and lying in xy plane in shown in figure. The plot of $\hat{j}$ component of magnetic field (By) at a distance 'a' (less than radius of the coil) and on yz plane vs z coordinate look like



(1) 

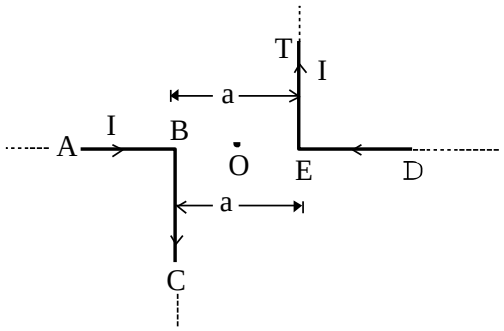
(2) 

(3) 

(4) 

PME317

17. The magnitude of magnetic induction at mid-point O due to current arrangement as shown in Fig will be :



(1) $\frac{\mu_0 I}{2\pi a}$ (2) 0

(3) $\frac{\mu_0 I}{4\pi a}$ (4) $\frac{\mu_0 I}{\pi a}$

PME318

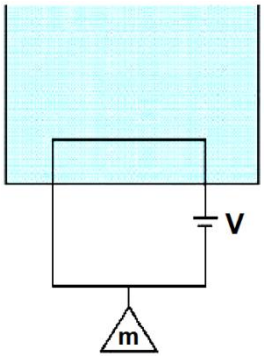
18. The electric current in a circular coil of four turns produces a magnetic induction 32 T at its centre. The coil is unwound and is rewound into a circular coil of single turn, the magnetic induction at the centre of the coil by the same current will be :

(1) 8T (2) 4T (3) 2T (4) 16T

PME319

19. A massless square loop, of wire of resistance $10\,\Omega$. supporting a mass of 1g. hangs vertically with one of its sides in a uniform magnetic field of 10^3 G, directed outwards in the shaded region. A dc voltage V is applied to the loop. For what value of V. the magnetic force will exactly balance the weight of the supporting mass of 1g?

(If sides of the loop = 10 cm, g = 10 ms^{-2})

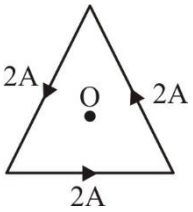


(1) $\frac{1}{10}$ V (2) 100 V

(3) 1 V (4) 10 V

PME320

20. As shown in the figure, a current of 2A flowing in an equilateral triangle of side $4\sqrt{3}$ cm. The magnetic field at the centroid O of the triangle is :



(Neglect the effect of earth's magnetic field.)

(1) $4\sqrt{3}\times10^{-4}$ T (2) $4\sqrt{3}\times10^{-5}$ T

(3) $\sqrt{3}\times10^{-4}$ T (4) $3\sqrt{3}\times10^{-5}$ T

PME321

21. A long conducting wire having a current I flowing through it, is bent into a circular coil of N turns. Then it is bent into a circular coil of n turns. The magnetic field is calculated at the centre of coils in both the cases. The ratio of the magnetic field in first case to that of second case is:

(1) N : n (2) $n^2 : N^2$

(3) $N^2 : n^2$ (4) n : N

PME322

22. A proton (mass m and charge +e) and an α-particle (mass 4m and charge +2e) are projected with the same kinetic energy at right angles to the uniform magnetic field. Which one of the following statements will be true

(1) The α-particle will be bent in a circular path with a small radius that for the proton

(2) The radius of the path of the α-particle will be greater than that of the proton

(3) The α-particle and the proton will be bent in a circular path with the same radius

(4) The α-particle and the proton will go through the field in a straight line

PME323

23. An electron is moving with a speed of 10^8 m/sec perpendicular to a uniform magnetic field of intensity B. Suddenly intensity of the magnetic field is reduced to B/2. The radius of the path becomes from the original value of r

(1) No change (2) Reduces to r / 2

(3) Increases to 2r (4) Stops moving

PME324

24. A proton and an α-particle enter a uniform magnetic field perpendicularly with the same speed. If proton takes 25 μ sec to make 5 revolutions, then the periodic time for the α-particle would be

(1) 50μ sec (2) 25μsec

(3) 10μ sec (4) 5μ sec

PME325

25. If a particle of charge 10^{-12} C moving along the $\hat{x}$ direction with a velocity 10^5 m/s experiences a force of 10^{-10} N in $\hat{y}$ direction due to magnetic field, then the minimum magnetic field is

(1) 6.25×10^3 T in $\hat{z}$ direction

(2) 10^{-15} T in $\hat{z}$ direction

(3) 6.25×10^{-3} T in $\hat{z}$ direction

(4) 10^{-3} T in $\hat{z}$ direction

PME326

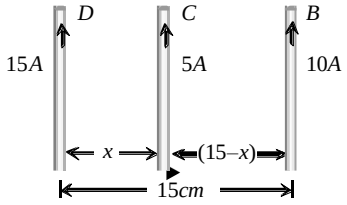
26. 3 A of current is flowing in a linear conductor having a length of 40 cm. The conductor is placed in a magnetic field of strength 500 gauss and makes an angle of 30° with the direction of the field. It experiences a force of magnitude

(1) 3×10^4 N (2) 3×10^2 N

(3) 3×10^{-2} N (4) 3×10^{-4} N

PME327

27. Three long, straight and parallel wires carrying currents are arranged as shown in the figure. The wire C which carries a current of 5.0 amp is so placed that it experiences no force. The distance of wire C from wire D is then

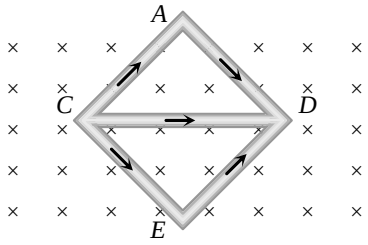


(1) 9 cm (2) 7 cm

(3) 5 cm (4) 3 cm

PME328

28. Same current i = 2A is flowing in a wire frame as shown in figure. The frame is a combination of two equilateral triangles ACD and CDE of side 1m. It is placed in uniform magnetic field B = 4T acting perpendicular to the plane of frame. The magnitude of magnetic force acting on the frame is



(1) 24 N (2) Zero (3) 16 N (4) 8 N

PME329

29. A charged particle is whirled in a horizontal circle on a frictionless table by attaching it to a string fixed at one point. If a magnetic field is switched on in the vertical direction, the tension in the string :-

(1) Will increase

(2) Will decrease

(3) Will remain the same

(4) May increase or decrease

PME330

30. A particle with a specific charge's' is fired with a speed v towards a wall at a distance d, perpendicular to the wall. What minimum magnetic field must exist in this region for the particle not to hit the wall :-

(1) $\frac{v}{sd}$ (2) $\frac{2v}{sd}$

(3) $\frac{v}{2sd}$ (4) $\frac{v}{4sd}$

PME331

31. A particle of charge per unit mass α is projected from origin with a velocity $\vec{v} = v_0\hat{i}$ in a uniform magnetic field $\vec{B} = -B_0\hat{k}$. If the particle passes through (0, y, 0) then y is equal to :-

(1) $-\frac{2v_0}{B_0\alpha}$ (2) $\frac{v_0}{B_0\alpha}$

(3) $\frac{2v_0}{B_0\alpha}$ (4) $-\frac{v_0}{B_0\alpha}$

PME332

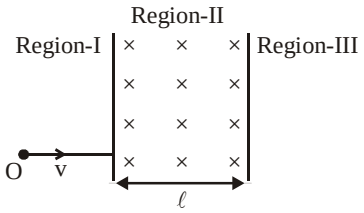
32. A charged particle enters in a magnetic field at right angles to the magnetic field. The field exists for a length equal to 1.5 times the radius of circular path of the particle. The particle will be deviated from its path by :-

(1) 90° (2) $\sin^{-1}\left(\frac{2}{3}\right)$

(3) 30° (4) 180°

PME333

33. A particle of mass m and charge q moving with velocity $v < \frac{qB\ell}{m}$ enters region-II normal to the boundary as shown in the figure. Region-II has a uniform magnetic field B perpendicular to the plane of paper. The particle will :-



(1) Enter the region-III
(2) Not enter the region-III
(3) Complete circular motion in region-II
(4) Move undeflected in region-II

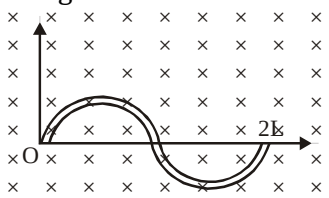
PME334

34. A small circular flexible loop of wire of radius R carries a current I. It is placed in a uniform magnetic field B. The tension in the loop will be doubled, if :-

(1) I is doubled (2) B is doubled
(3) R is doubled (4) All of these

PME335

35. A wire carrying a current i is placed in a uniform magnetic field 'B'. This wire is in the form of the curve $y = a \sin\left(\frac{\pi x}{L}\right)$ $0 \le x \le 2L$. Force acting on the wire is :-



(1) $\frac{iBL}{\pi}$ (2) $iBL\pi$ (3) $2iBL$ (4) Zero

PME336

36. The magnetic field existing in a region is given by $\vec{B} = B_0\left(1+\frac{x}{\ell}\right)\hat{k}$. A square loop of edge ℓ and carrying a current i, is placed with its edge parallel to the x-y axes. Find the magnitude of the net magnetic force experienced by the square loop :-

(1) $\frac{1}{2}iB_0\ell$ (2) Zero (3) $iB_0\ell$ (4) $2iB_0\ell$

PME337

37. Suppose an isolated north pole is kept at the centre of a circular loop carrying a electric current i. The magnetic field due to the north pole at a point on the periphery of the wire is B. The radius of the loop is a. The force on the wire is :-

(1) Nearly 2 π a i B perpendicular to the plane of loop
(2) 2 π a i B perpendicular to the plane of loop
(3) π a i B along the axis of the loop
(4) Zero

PME338

38. There are 50 turns of a wire in every cm length of a long solenoid. If 4 ampere current is flowing in the solenoid, the approximate value of magnetic field along its axis at an internal point and at one end will be respectively

(1) 12.6×10^{-3} Weber/m², 6.3×10^{-3} Weber/m²
(2) 12.6×10^{-3} Weber/m², 25.1×10^{-3} Weber/m²
(3) 25.1×10^{-3} Weber/m², 12.6×10^{-3} Weber/m²
(4) 25.1×10^{-5} Weber/m², 12.6×10^{-5} Weber/m²

PME339

39. The current in the windings on a toroid is 2.0A. There are 400 turns and the mean circumferential length is 40cm. If the inside magnetic field is 1.0T, the relative permeability is near to :-

(1) 100 (2) 200
(3) 300 (4) 400

PME340

40. A long straight wire carries a current. Can Ampere's circuital law be used to find magnetic field on a loop that doesn't enclose the wire: -

(1) Yes, if loop is circular

(2) Yes, if loop is rectangular

(3) Yes, if loop is of irregular shape

(4) No

PME341

ANSWER KEY

1.	(1)	**2.**	(1)	**3.**	(4)	**4.**	(1)	**5.**	(4)
6.	(2)	**7.**	(2)	**8.**	(2)	**9.**	(2)	**10.**	(3)
11.	(3)	**12.**	(1)	**13.**	144	**14.**	(1)	**15.**	(3)
16.	(3)	**17.**	(4)	**18.**	(3)	**19.**	(4)	**20.**	(4)
21.	(3)	**22.**	(3)	**23.**	(3)	**24.**	(3)	**25.**	(4)
26.	(3)	**27.**	(1)	**28.**	(1)	**29.**	(4)	**30.**	(1)
31.	(3)	**32.**	(4)	**33.**	(2)	**34.**	(4)	**35.**	(3)
36.	(3)	**37.**	(2)	**38.**	(3)	**39.**	(4)	**40.**	(4)

Magnetic Effect of Current and Magnetism-2

1. Surface charge density on a ring of radius a and width d is σ as shown in the figure. It rotates with frequency f about its own axis. Assume that the charge is only on outer surface. The magnetic field induction at centre is (Assume that d << a) :-



(1) $\pi \mu_0 f \sigma d$ (2) $\mu_0 f \sigma d$

(3) $2\pi \mu_0 f \sigma d$ (4) $\frac{\pi^2}{2\mu_0} f \sigma d$

PME342

2. A bar magnet having a magnetic moment of 2.0×10^5 JT^{-1}, is placed along the direction of uniform magnetic field of magnitude $B = 14 \times 10^{-5}$ T. The work done in rotating the magnet slowly through 60° from the direction of field is :

(1) 14 J (2) 8.4 J (3) 4 J (4) 1.4 J

PME343

3. The susceptibility of a paramagnetic material is 99. The permeability of the material in Wb/A-m is : [Permeability of free space $\mu_0 = 4\pi \times 10^{-7} Wb/A-m$]

(1) $4\pi \times 10^{-7}$ (2) $4\pi \times 10^{-4}$

(3) $4\pi \times 10^{-5}$ (4) $4\pi \times 10^{-6}$

PME344

4. A solenoid of 1200 turns is wound uniformly in a single layer on a glass tube 2 m long and 0.2 m in diameter. The magnetic intensity at the center of the solenoid when a current of 2 A flows through it is:

(1) 2.4×10^3 A m^{-1} (2) 1.2×10^3 A m^{-1}

(3) 1 A m^{-1} (4) 2.4×10^{-3} A m^{-1}

PME345

5. The magnetic moments associated with two closely wound circular coils A and B of radius r_A = 10 cm and r_B = 20 cm respectively are equal if: (Where N_A, I_A and N_B, I_B are number of turns and current of A and B respectively)

(1) $2N_A I_A = N_B I_B$ (2) $N_A = 2N_B$

(3) $N_A I_A = 4N_B I_B$ (4) $4N_A I_A = N_B I_B$

PME346

6. The magnetic moment of a current carrying loop is 2.1×10^{-25} amp × m^2. The magnetic field at a point on its axis at a distance of 1Å is

(1) 4.2×10^{-2} weber/m^2

(2) 4.2×10^{-3} weber/m^2

(3) 4.2×10^{-4} weber/m^2

(4) 4.2×10^{-5} weber/m^2

PME347

7. In a hydrogen atom, an electron moves in a circular orbit of radius 5.2×10^{-11}m and produces a magnetic induction of 12.56 T at its nucleus. The current produced by the motion of the electron will be

(Given $\mu_0 = 4\pi \times 10^{-7}$ Wb/A–m)

(1) 6.53×10^{-3} ampere

(2) 13.25×10^{-10} ampere

(3) 9.6×10^6 ampere

(4) 1.04×10^{-3} ampere

PME348

8. The earth's magnetic induction at a certain point is 7×10^{-5} Wb/m^2. This is to be annulled by the magnetic induction at the centre of a circular conducting loop of radius 5 cm. The required current in the loop is

(1) 0.56 A (2) 5.6 A

(3) 0.28 A (4) 2.8 A

PME349

9. Two wires of same length are shaped into a square and a circle. If they carry same current, ratio of the magnetic moment is

(1) 2 : π (2) π : 2 (3) π : 4 (4) 4 : π

PME350

10. A circular coil of radius 4 cm has 50 turns. In this coil a current of 2 A is flowing. It is placed in a magnetic field of 0.1 weber/m^2. The amount of work done in rotating it through 180° from its stable equilibrium position will be

(1) 0.1 J (2) 0.2 J

(3) 0.4 J (4) 0.8 J

PME351

11. Magnetostatics screening or shielding can be created by :-
(1) Super conductor
(2) Soft iron ring
(3) Both (1) & (2)
(4) Neither (1) nor (2)

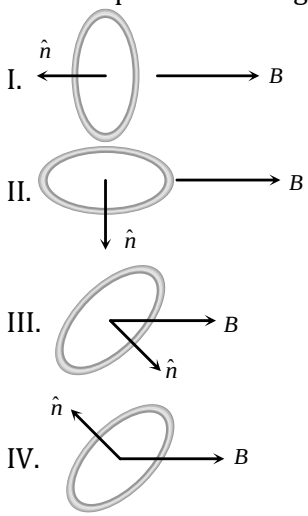
PME352

12. Current i is carried in a wire of length L. If the wire is turned into a circular coil, the maximum magnitude of torque in a given magnetic field B will be

(1) $\frac{LiB^2}{2}$ (2) $\frac{Li^2B}{2}$ (3) $\frac{L^2iB}{4\pi}$ (4) $\frac{Li^2B}{4\pi}$

PME353

13. A current carrying loop is placed in a uniform magnetic field in four different orientations, I,II, III & IV arrange them in the decreasing order of potential Energy



(1) I > III > II > IV (2) I > II >III > IV
(3) I > IV > II > III (4) III > IV > I > II

PME354

14. A rigid circular loop of radius R and mass m lies in the x-y plane on a flat table and has a current i flowing in it. At this particular place the earth's magnetic field is $\vec{B}=B_x\hat{i}+B_z\hat{k}$. The value of i, so that the loop starts tilting is :-

(1) $\frac{mg}{\pi R\sqrt{B_x^2+B_2^2}}$ (2) $\frac{mg}{\pi RB_x}$

(3) $\frac{mg}{\pi RB_z}$ (4) $\frac{mg}{\pi R\sqrt{B_xB_z}}$

PME355

15. Two short bar magnets of magnetic moments 'M' each are arranged at the opposite corners of a square of side 'd', such that their centres coincide with the corners and their axes are parallel to one side of the square. If the like poles are in the same direction, the magnetic induction at any of the other corners of the square is :-

(1) $\frac{\mu_0}{4\pi}\frac{M}{d^3}$ (2) $\frac{\mu_0}{4\pi}\frac{2M}{d^3}$

(3) $\frac{\mu_0}{2\pi}\frac{M}{d^3}$ (4) $\frac{\mu_0}{2\pi}\frac{2M}{d^3}$

PME356

16. A magnetising field of 1600 Am^{-1} produces a magnetic flux 2.4×10^{-5} weber in an iron bar of cross-sectional area 0.2 cm^2. The susceptibility of the iron bar will be :-

(1) 1788 (2) 1192
(3) 596 (4) 298

PME357

ANSWER KEY

1.	(1)	**2.**	(1)	**3.**	(3)	**4.**	(2)	**5.**	(3)
6.	(1)	**7.**	(4)	**8.**	(2)	**9.**	(3)	**10.**	(1)
11.	(3)	**12.**	(3)	**13.**	(3)	**14.**	(2)	**15.**	(1)
16.	(3)								

Electromagnetic Induction

1. A conducting ring of radius r is placed perpendicularly inside a time varying magnetic field given by $B = B_0 + \alpha t$. B_0 and α are positive constants, E.m.f. induced in the ring is :-

(1) $-\pi\alpha r$ (2) $-\pi\alpha r^2$

(3) $-\pi\alpha^2 r^2$ (4) $-\pi\alpha^2 r$

PEI196

2. If the rotational velocity of dynamo armature is four times, then the induced emf of the dynamo will be :-

(1) Four times (2) Eight times

(3) Half (4) None of these

PEI197

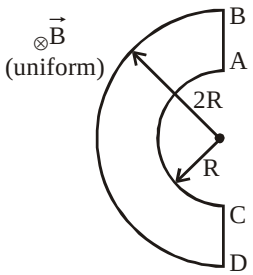
3. A current of 1A and frequency n_1 is passed through a coil subsequently the frequency is made n_2. If ε_1 and ε_2 be the induced peak emf in the coil, in the two cases then $\frac{\varepsilon_1}{\varepsilon_2}$ will be equal to :-

(1) $n_2^2 : n_1^2$ (2) $n_1^2 : n_2^2$

(3) $n_1 : n_2$ (4) $n_2 : n_1$

PEI198

4. If given arrangement is moving towards left with speed v, then potential difference between B and D and current in the loop are respectively :-

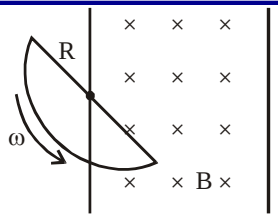


(1) BvR and non-zero (2) 2BvR and zero

(3) 4BvR and non-zero (4) 4BvR and zero

PEI199

5. A semicircular loop of radius R is rotated with an angular velocity ω. perpendicular to the plane of a uniform magnetic field B as shown in the figure. Emf induced in the loop is :



(1) $B\omega R^2$ (2) $\frac{1}{2} B\omega R^2$

(3) $\frac{3}{2} B\omega R^2$ (4) $\frac{1}{4} B\omega R^2$

PEI200

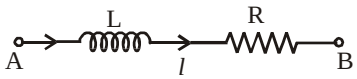
6. A conducting ring of radius b is placed coaxially in a long solenoid of radius a(b < a) having n turns per unit length. A current $i = i_0 \cos \omega t$ flows through the solenoid. The induced emf in the ring is :-

(1) Zero (2) $\mu_0 n i_0 \pi a^2 \omega$

(3) $\mu_0 n i_0 \pi a^2 \cos \omega t$ (4) $\mu_0 n i_0 \pi b^2 \omega \sin \omega t$

PEI201

7. In an A.C. sub-circuit as shown in figure, the resistance R = 0.2 Ω. At a certain instant $V_A - V_B$ = 0.5 V, I = 0.5 A, and current is increasing at the rate of $\frac{\Delta I}{\Delta t}$ = 8 A/s. The inductance of the coil is

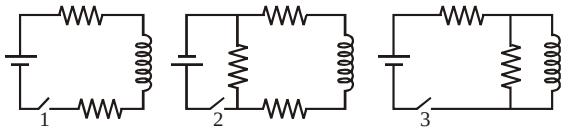


(1) 0.01 H (2) 0.02 H

(3) 0.05 H (4) 0.5 H

PEI202

8.



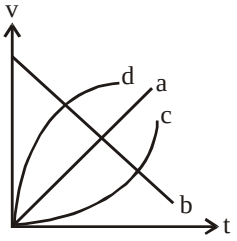
The figure shows three circuits with identical batteries, inductors and resistances. Rank the circuits according to the currents through the battery just after the switch is closed, greatest first.

(1) $I_2 > I_3 > I_1$ (2) $I_2 > I_1 > I_3$

(3) $I_1 > I_2 > I_3$ (4) $I_1 > I_3 > I_2$

PEI203

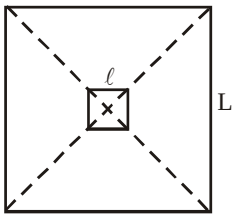
9. A bar magnet is made to fall through a long vertical copper tube. The speed (v) of the magnet as a function of time (t) is best represented by :-



(1) a (2) b (3) c (4) d

PEI204

10. A small square loop of wire of side *l* is placed inside a large square loop of wire of side (L >> *l*). The loops are coplanar and their centres coincide. The mutual inductance of the system is :

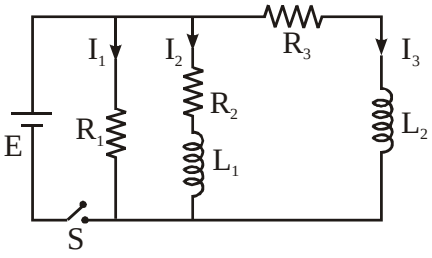


(1) $\frac{2\sqrt{2}\mu_0 L}{\pi\ell}$ (2) $\frac{2\sqrt{2}\mu_0 L^2}{\pi\ell}$

(3) $\frac{2\sqrt{2}\mu_0 \ell}{\pi L}$ (4) $\frac{2\sqrt{2}\mu_0 \ell^2}{\pi L}$

PEI205

11. Figure shows an L-R circuit. When the switch S is closed, the current through resistor R_1, R_2 and R_3 are I_1, I_2 and I_3 respectively. The value of I_1, I_2 and I_3 at t = 0 s is :



(1) $I_1 = I_2 = I_3 = 0$

(2) $I_1 = \frac{E}{R_1}$, $I_2 = I_3 = 0$

(3) $I_1 = 0$, $I_2 = \frac{E}{R_2}$, $I_3 = \frac{E}{R_3}$

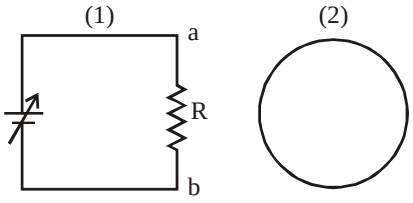
(4) $I_1 = \frac{E}{R_1}$, $I_2 = \frac{E}{R_2 + L_1}$, $I_3 = \frac{E}{R_3 + L_3}$

PEI206

12. If magnetic field passing through a coil of area 0.1 m^2 is changing according to the equation $B = 10 \sin\left(\frac{2\pi t}{3}\right)$ tesla. Find the magnitude of induced emf at t = 0.5s

PEI207

13. The emf of the battery in loop-(1) is increased. Find the direction of induced current in the loop-(2).



PEI208

14. In LR circuit the current increases to one fourth of its maximum value in 4sec than the time constant of circuit is :-

(1) $2 \log_e 2$ (2) $\frac{4}{\log_e 2}$

(3) $\frac{4}{\log_e (4/3)}$ (4) $\frac{\log_e 2}{2}$

PEI209

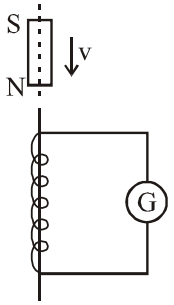
15. A small cube of volume 1 mm^3 placed at the centre of circular loop of radius of 100 cm, carrying a current 1A. The magnetic energy stored in side cube is :-

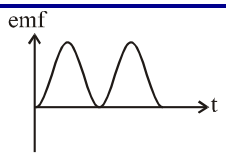
(1) $\frac{\pi}{2} \times 10^{16}$ J (2) $2\pi \times 10^{-7}$ J

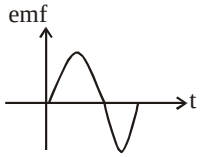
(3) $\frac{\pi}{2} \times 10^{-16}$ J (4) $\frac{\pi}{2} \times 10^{-7}$ J

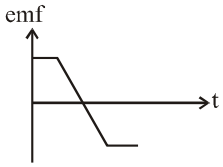
PEI210

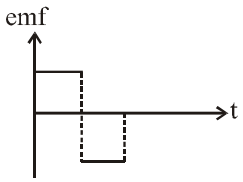
16. A bar magnet released from rest in the effect of gravity along vertical axis of coil then which of the following variation is correct regarding induced emf with respect to time when it passes through the coil :-



(1) 

(2) 

(3) 

(4) 

PEI211

17. Two coils of self inductance L_1 and L_2 are connected in series combination having mutual inductance of the coils as M. The equivalent self inductance of the combination will be :

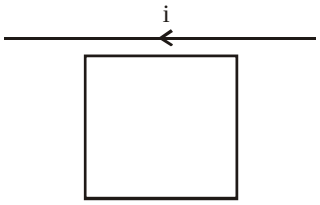


(1) $\frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{M}$ (2) $L_1 + L_2 + M$

(3) $L_1 + L_2 + 2M$ (4) $L_1 + L_2 - 2M$

PEI212

18. Current i flows through a long wire. A square loop made of conducting wire held below the wire in the same vertical plane is released and allowed to fall under gravity. If acceleration of the loop is a, then :-



(1) A current is induced in anticlockwise direction and a > g

(2) A current is induced in the clockwise direction and a < g

(3) A current is induced in anticlockwise direction and a < g

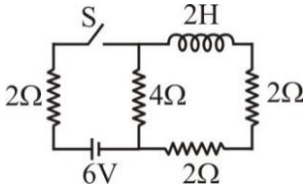
(4) No current is induced in the loop and a = g

PEI213

19. A conducting circular loop is placed in X – Y plane in presence of magnetic field $\vec{B}=\left(3t^3\hat{j}+3t^2\hat{k}\right)$ in SI unit. If the radius of the loop is 1m, the induced emf in the loop, at time, t = 2s is nπV. The value of n is……..

PEI214

20. For the given circuit the current through battery of 6 V just after closing the switch 'S' will be ……. A.



PEI215

21. Match the List-I with List-II.

	List I		**List II**
A.	AC generator	I.	Presence of both L and C
B.	Transformer	II.	Electromagnetic Induction
C.	Resonance phenomenon to occur	III.	Quality factor
D.	Sharpness of resonance	IV.	Mutual Inductance

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-I, D-III

(2) A-II, B-I, C-III, D-IV

(3) A-II, B-IV, C-I, D-III

(4) A-IV, B-III, C-I, D-II

PEI216

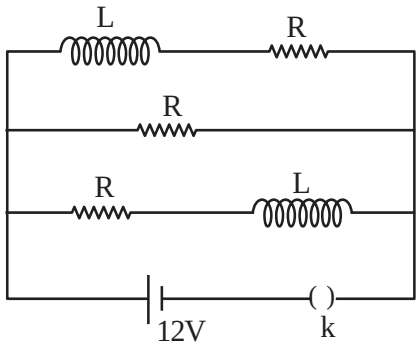
22. A conducting loop of radius $\frac{10}{\sqrt{\pi}}$ cm is placed perpendicular to a uniform magnetic field of 0.5T. The magnetic field is decreased to zero in 0.5 s at a steady rate. The induced emf in the circular loop at 0.25 s is:

(1) emf = 1 mV (2) emf = 10 mV

(3) emf = 100 mV (4) emf = 5 mV

PEI217

23. Three identical resistors with resistance R = 12Ω and two identical inductors with sell inductance L=5 mH are connected to an ideal battery with emf of 12 V as shown in figure. The current through the battery long after the switch has been closed will be_____A.



PEI218

24. Two coils have self inductance L_1 = 4mH and L_2 = 1 mH respectively. The currents in the two coils are increased at the same rate. At a certain instant of time both coils are given the same power. If I_1 and I_2 are the currents in the two coils, at that instant of time respectively, then the value of $\left(\frac{I_1}{I_2}\right)$ is :-

(1) $\frac{1}{8}$ (2) $\frac{1}{4}$ (3) $\frac{1}{2}$ (4) 1

PEI219

25. In an ac generator, a rectangular coil of 100 turns each having area 14×10^{-2} m^2 is rotated at 360 rev/min about an axis perpendicular to a uniform magnetic field of magnitude 3.0 T. The maximum value of the emf produced will be _____ V. $\left(\text{Take } \pi = \frac{22}{7}\right)$

PEI220

ANSWER KEY

1.	(2)	**2.**	(1)	**3.**	(3)	**4.**	(4)	**5.**	(2)
6.	(4)	**7.**	(3)	**8.**	(1)	**9.**	(4)	**10.**	(4)
11.	(2)	**12.**	$\frac{\pi}{3}$ volt	**13.**	Clockwise	**14.**	(3)	**15.**	(3)
16.	(2)	**17.**	(4)	**18.**	(3)	**19.**	12	**20.**	1
21.	(3)	**22.**	(2)	**23.**	3	**24.**	(2)		
25.	1584								

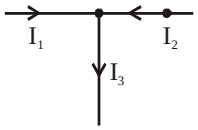
Alternating Current

1. The voltage supplied to a circuit is given by $V = V_0t^{3/2}$, where t is time in second. Find the RMS value of voltage for the period t = 0 to t = 1s :-

(1) $\frac{V_0}{2}$ (2) V_0 (3) $\frac{3V_0}{2}$ (4) $2V_0$

PAC170

2. If current $I_1 = 3A \sin \omega t$ and $I_2 = 4A \cos \omega t$, then I_3 is :-



(1) 5A sin (ωt + 53°) (2) 5A sin (ωt + 37°)
(3) 5A sin (ωt + 45°) (4) 5A sin (ωt + 30°)

PAC171

3. What is the approximate peak value of an alternating current producing four times the heat produced per second by a steady current of 2 A in a resistor :-

(1) 2.8 A (2) 4.0 A (3) 5.7 A (4) 8.0 A

PAC172

4. An electric lamp marked 100 volt d.c. consumes a current of 10A. If it is connected to a 200 volt, 50 Hz AC mains. Calculate the inductance of choke (inductor coil) required:-

(1) 5.5×10^{-2} H (2) 5.5×10^{-1} H
(3) 11.0 H (4) 11.0×10^{-2} H

PAC173

5. A long solenoid connected to a 12 volt DC source passes a steady current of 2A. When the solenoid is connected to an AC source of 12 volt at 50 Hz, the current flowing is 1A. Calculate inductance of the solenoid :-

(1) 33 mH (2) 66 mH
(3) 16.5 mH (4) 40 mH

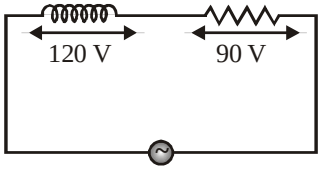
PAC174

6. When an electric device X is connected to a 220 volt, 50 hertz a.c. supply, the current is 0.5 A, and is in same phase as the applied voltage. When another device Y is connected to the same supply, the electric current is again 0.5 A, but it leads the potential difference by $\frac{\pi}{2}$. When X and Y are connected in series across the same source, what will be the current?

(1) 0.35 A (2) 0.55 A
(3) 1.35 A (4) 2.5 A

PAC175

7. If current in the circuit is 3A, then impedance of the circuit shown in the figure is :-



(1) 100 Ω (2) 50 Ω
(3) 30 Ω (4) 40 Ω

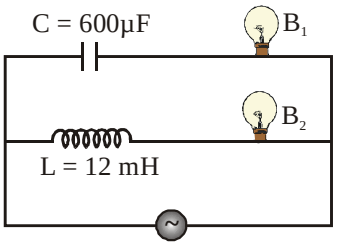
PAC176

8. A direct current of 10 A is superimposed on an alternating current i = 40 cos ωt(A) flowing through a wire. The effective value of the resulting current will be :-

(1) $10\sqrt{2}$ A (2) $20\sqrt{2}$ A
(3) $20\sqrt{3}$ A (4) 30 A

PAC177

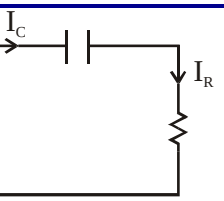
9. In the circuit shown in figure, if both the bulbs B_1 and B_2 identical then :-

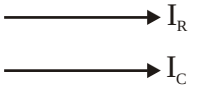


(1) Both bulbs have equal brightness
(2) B_2 will be brighter than B_1
(3) At very high frequency of source brightness of B_1 is more than B_2.
(4) At the very low frequency of source the brightness of B_1 is more than B_2.

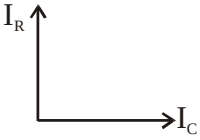
PAC178

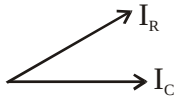
10. Given figure shows a source of alternating voltage connected to a capacitor and a resistor. Which of the following phasor diagrams correctly describes the phase relationship between I_C, the current between the source and the capacitor and I_R, the current in resistor ?



(1) 

(2) 

(3) 

(4) 

PAC179

11. An AC voltage is represented by

$E = 575\sqrt{5}\cos(75\pi)t$

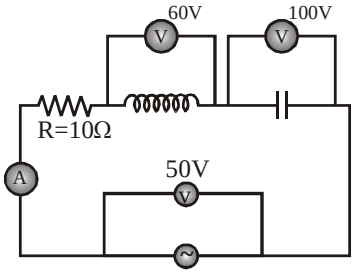
How many times will the current become zero in 1 second?

(1) 75 times (2) 50 times

(3) 100 times (4) 575 times

PAC180

12. In the circuit shown in figure the current measured by ammeter is :-

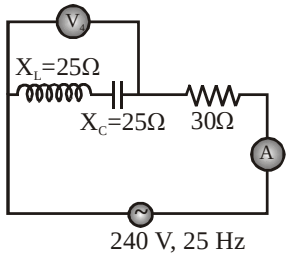


(1) 1 A (2) 2 A

(3) 3 A (4) Zero

PAC181

13. In the circuit shown in figure neglecting source resistance, the voltmeter and ammeter reading will respectively be :-

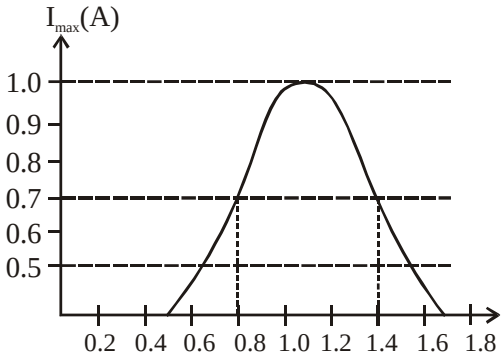


(1) 0 V, 3 A (2) 150 V, 3 A

(3) 150 V, 6 A (4) 0 V, 8 A

PAC182

14. For series L-C-R circuit the plot of I_{max} versus ω is shown in the figure. The bandwidth will be:



(1) 0.2 rad s^{-1}

(2) 0.3 rad/sec

(3) 0.6 rad/sec

(4) 0.5 rad/sec

PAC183

15. Match List-I with List-II

List-I		List-II	
(A)	AC generator	(I)	Detects the presence of current in the circuit
(B)	Galvanometer	(II)	Converts mechanical energy into electrical energy
(C)	Transformer	(III)	Works on the principle of resonance in AC circuit
(D)	Metal detector	(IV)	Changes an alternating voltage for smaller or greater value

Choose the **correct answer** from the options given below :-

(1) (A)–(II), B–(I), (C)–(IV), (D)–(III)

(2) (A)–(II), B–(I), (C)–(III), (D)–(IV)

(3) (A)–(III), B–(IV), (C)–(II), (D)–(I)

(4) (A)–(III), B–(I), (C)–(II), (D)–(IV)

PAC184

16. If L, C and R are the self inductance, capacitance and resistance respectively, which of the following does not have the dimension of time?

(1) RC (2) $\frac{L}{R}$

(3) $\sqrt{LC}$ (4) $\frac{L}{C}$

PAC185

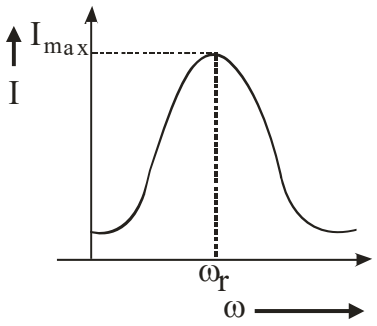
17. For a series LCR circuit, I vs ω curve is shown:

(a) To the left of ω_r, the circuit is mainly capacitive.

(b) To the left of ω_r, the circuit is mainly inductive.

(c) At ω_r, impedance of the circuit is equal to the resistance of the circuit.

(d) At ω_r, impedance of the circuit is 0.



Choose the **most appropriate** answer from the options given below :

(1) (a) and (d) only

(2) (b) and (d) only

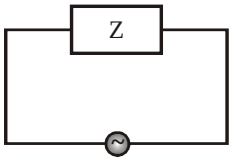
(3) (a) and (c) only

(4) (b) and (c) only

PAC186

18. In a box Z of unknown elements, L or R or any other combination, an ac voltage $E = E_0 \sin(\omega t + \phi)$ is applied and current in the circuit was found to be $I = I_0 \sin\left(\omega t + \phi + \frac{\pi}{4}\right)$.

The unknown element in the box may be :-



(1) Only capacitor

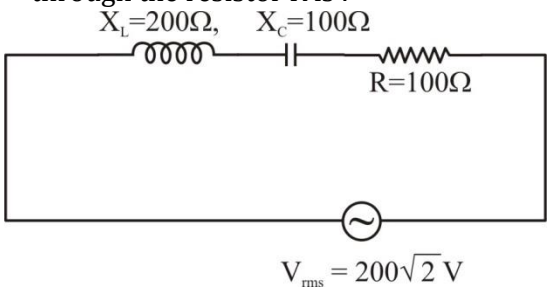
(2) inductor and resistor both

(3) Either capacitor, resistor and inductor or only capacitor and resistor

(4) Only resistor

PAC187

19. In the given circuit, rms value of current (I_{rms}) through the resistor R is :



(1) 2A (2) $\frac{1}{2}$A

(3) 20 A (4) $2\sqrt{2}$A

PAC188

20. A series LCR circuit consists of R=80Ω. X_L=100Ω, and X_C=40Ω. The input voltage is 2500 cos(100 πt) V. The amplitude of current, in the circuit, is_____A.

PAC189

ANSWER KEY

1.	(1)	**2.**	(1)	**3.**	(3)	**4.**	(1)	**5.**	(1)
6.	(1)	**7.**	(2)	**8.**	(4)	**9.**	(3)	**10.**	(1)
11.	(1)	**12.**	(3)	**13.**	(4)	**14.**	(3)	**15.**	(1)
16.	(4)	**17.**	(3)	**18.**	(3)	**19.**	(1)	**20.**	25

Electromagnetic Waves

1. A free proton is placed in the path of a plane electromagnetic wave. The proton will start moving: -
(1) In the direction of propagation of EM wave
(2) Along the magnetic field
(3) Along the electric field
(4) In the plane containing the magnetic field and the direction of propagation

PEW056

2. Which of the following statements is incorrect about electromagnetic waves?
(1) Are transverse in nature
(2) Are produced by accelerating charges
(3) Travel with the same speed in all media
(4) Travel in free space with the speed of light

PEW057

3. In electromagnetic wave the phase difference between electric and magnetic field vectors is
(1) zero (2) $\pi/2$ (3) π (4) $\pi/4$

PEW058

4. Total energy density of electromagnetic waves in vacuum in given by the relation :-
(1) $\frac{1}{2}\cdot\frac{E^2}{\varepsilon_0}+\frac{B^2}{2\mu_0}$
(2) $\frac{1}{2}\varepsilon_0E^2+\frac{1}{2}\mu_0B^2$
(3) $\frac{E^2+B^2}{c}$
(4) $\frac{1}{2}\varepsilon_0E^2+\frac{B^2}{2\mu_0}$

PEW059

5. In free space an electromagnetic wave that propagates in the +z direction with an electric field strength of 1 V/m pointing in the +y direction. Then the direction and magnitude of the magnetic field is :-
(1) 3.33×10^{-9} T in (–y) direction
(2) 3.33×10^{-9} T in (–x) direction
(3) 3.33×10^{-9} T in (+x) direction
(4) 9.99×10^{-7} T in (–x) direction

PEW060

6. If electric field in a non-magnetic medium for EM wave is given by $\vec{E}=100\sin(10^8t+2z)\hat{j}$ V/m then $\vec{H}$ is (in S.I.) :-
(1) $\frac{50}{\pi}\sin(10^8t+2z)\hat{i}$ (2) $\frac{5}{\pi}\sin(10^8t+2z)\hat{i}$
(3) $20\sin(10^8t+2z)\hat{j}$ (4) $\frac{25}{\pi}\sin(10^8t+2z)\hat{j}$

PEW061

7. The velocity of electromagnetic wave is parallel to :-
(1) $\vec{B}\times\vec{E}$ (2) $\vec{E}\times\vec{B}$ (3) $\vec{E}$ (4) $\vec{B}$

PEW062

8. Electromagnetic wave of intensity 1400 W/m^2 falls normally on a surface of area 1.5 m^2 is completely absorbed by it. Find out force exerted by beam.
(1) 14×10^{-5} N (2) 14×10^{-6} N
(3) 7×10^{-5} N (4) 7×10^{-6} N

PEW063

9. The electric field in an electromagnetic wave is given by $E=56.5\sin\omega(t-x/c)NC^{-1}$. Find the intensity of the wave if it is propagating along x-axis in the free space.
(Given $\varepsilon_0=8.85\times10^{-12}C^2N^{-1}m^{-2}$)
(1) 5.65 Wm^{-2} (2) 4.24 Wm^{-2}
(3) 1.9×10^{-7} Wm^{-2} (4) 56.5 Wm^{-2}

PEW064

10. The electromagnetic waves travel in a medium at a speed of 2.0×10^8 m/s. The relative permeability of the medium is 1.0. The relative permittivity of the medium will be:
(1) 2.25 (2) 4.25 (3) 6.25 (4) 8.25

PEW065

11. Light wave travelling in air along x-direction is given by $E_y = 540\sin\pi\times10^4(x-ct)$ Vm^{-1}. Then, the peak value of magnetic field of wave will be (Given $c = 3\times10^8$ ms^{-1})
(1) 18×10^{-7} T (2) 54×10^{-7} T
(3) 54×10^{-8} T (4) 18×10^{-8} T

PEW066

12. Match List – I with List – II

List – I	List - II
(a) UV rays	(i) Diagnostic tool in medicine
(b) X-rays	(ii) Water purification
(c) Microwave	(iii) Communication, Radar
(d) Infrared wave	(iv) Improving visibility in foggy days

Choose the correct answer from the options given below :

(1) (a)–(iii), (b)-(ii), (c)-(i), (d)-(iv)

(2) (a)–(ii), (b)-(i), (c)-(iii), (d)-(iv)

(3) (a)–(ii), (b)-(iv), (c)-(iii), (d)-(i)

(4) (a)–(iii), (b)-(i), (c)-(ii), (d)-(iv)

PEW067

13. If $\vec{E}$ and $\vec{K}$ represent electric field and propagation vectors of the EM waves in vacuum, then magnetic field vector is given by : (ω - angular frequency) :

(1) $\frac{1}{\omega}\left(\vec{K}\times\vec{E}\right)$ (2) $\omega\left(\vec{E}\times\vec{K}\right)$

(3) $\omega\left(\vec{K}\times\vec{E}\right)$ (4) $\vec{K}\times\vec{E}$

PEW068

14. Displacement current goes through the gap between the plates of a capacitor when the charge of the capacitor :-

(a) does not change (b) change with time

(c) increases (d) decreases

(1) (a), (c) and (d) (2) (a), (b) and (d)

(3) (b), (c) and (d) (4) (a), (b) and (c)

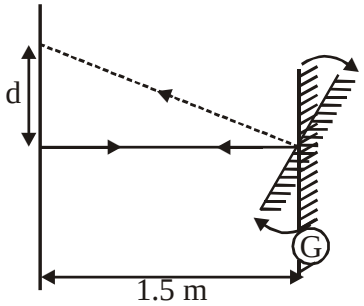
PEW069

ANSWER KEY

1.	(3)	**2.**	(3)	**3.**	(1)	**4.**	(4)	**5.**	(2)
6.	(2)	**7.**	(2)	**8.**	(4)	**9.**	(2)	**10.**	(1)
11.	(1)	**12.**	(2)	**13.**	(1)	**14.**	(3)		

Ray Optics and Optical Instruments

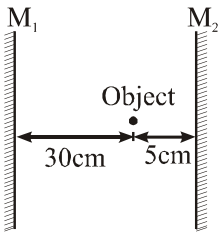
1. Light incident normally on a plane mirror attached to a galvanometer coil retraces backwards as shown in figure. A current in the coil produces a deflection of 3.5° of the mirror. What is the displacement of reflected spot of light on a screen placed 1.5 m away ?



(1) 10.5 cm (2) 18.31 cm
(3) 15 cm (4) 3 cm

PRI221

2.



Position of 5th image of given object in mirror M_2 is :-
(1) 65 cm from mirror M_2
(2) 145 cm from mirror M_2
(3) 170 cm from mirror M_2
(4) 135 cm from mirror M_2

PRI222

3. A point object approaches a plane mirror with a speed of 10 ms^{-1}, while the image recedes away from the mirror with a speed of 6 ms^{-1}. The direction and magnitude of the velocity of mirror is :-
(1) towards the object, 8 ms^{-1}
(2) towards the image, 6 ms^{-1}
(3) away from the object, 8 ms^{-1}
(4) away from the object, 2 ms^{-1}

PRI223

4. The focal length of a concave mirror is 50cm. Where an object be placed so that its image is two times magnified and inverted :-
(1) 75 cm (2) 72 cm
(3) 63 cm (4) 50 cm

PRI224

5. An object is placed at 20 cm from a convex mirror of focal length 10 cm. The image formed by the mirror is :-
(1) Real and at 10 cm from the mirror
(2) Virtual and at 10 cm from the mirror
(3) Virtual and at 20/3 cm from the mirror
(4) Real and at 20/3 cm from the mirror

PRI225

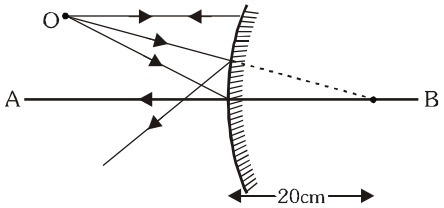
6. A small candle 2.5 cm in size is placed 27 cm in front of a concave mirror of radius of curvature 36 cm. At what distance from the mirror should a screen be placed in order to receive a sharp image. What is size of image :-

[NCERT]

(1) 54, 5 (2) 54, 10
(3) 30, 5 (4) 30, 10

PRI226

7. In the figure a convex mirror of radius of curvature 20 cm is shown. An object O is placed in front of this mirror. Its ray diagram is shown. How many mistakes are there in the diagram? (AB is its principal axis)



(1) 3 (2) 2 (3) 1 (4) 0

PRI227

8. A point object is placed 60 cm from pole of a concave mirror of focal length 10 cm on the principal axis. If object is shifted 1 mm towards the mirror along principal axis. Find the shift in image?

(1) $\frac{1}{25}$ mm (2) $\frac{1}{50}$ mm
(3) $\frac{1}{100}$ mm (4) $\frac{1}{125}$ mm

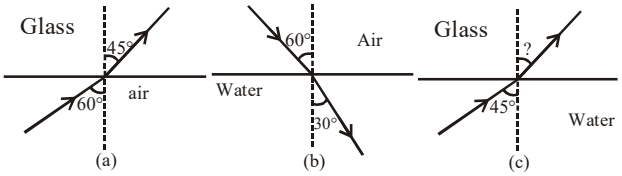
PRI228

9. A transparent cube of 15 cm edge contains a small air bubble. Its apparent depth when viewed through one face is 6 cm and when viewed through opposite face is 4 cm. The refractive index of material of cube is :-
(1) 2.0 (2) 1.5 (3) 1.6 (4) 2.5

PRI229

10. Figure (a) and (b) show refraction of a ray in air incident at 60° with the normal to a glass air and water air interface, respectively. What is the angle of refraction in glass when the angle of incidence in water is 45° with the normal to a water-glass interface figure (c):

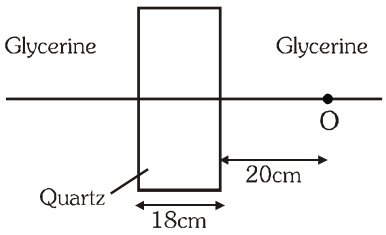
[NCERT]



(1) 38° (2) 60° (3) 90° (4) 52°

PRI230

11. Given that, velocity of light in quartz is 1.5×10^8 m/s and velocity of light in glycerine is $\frac{9}{4}\times 10^8$ m/s. Now a slab made of quartz is placed in glycerine as shown. What is the shift produced by slab, if seen from other side of slab.



(1) 6 cm (2) 3.55 cm
(3) 9 cm (4) 2 cm

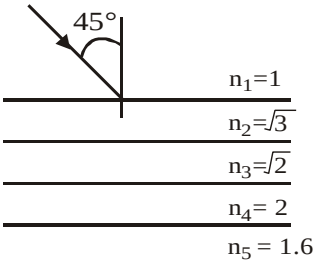
PRI231

12. How much water would be filled in a container of height 21 cm, so that it appears half filled to the observer when viewed from the top of the container? (μ_w=4/3)

(1) 8 cm (2) 10.5 cm
(3) 12 cm (4) 14 cm

PRI232

13. In the figure shown the angle made by the light ray with the normal in the medium of refractive index $\sqrt{2}$ is:



(1) 30° (2) 60°
(3) 90° (4) None of these

PRI233

14. A ray of light is incident from a rarer medium (μ = 1.5) into a denser medium. If the angle of incidence and refraction are respectively 60° and 45°, then calculate the refractive index of the denser medium.

(1) 1 (2) 1.73 (3) 1.5 (4) 1.84

PRI234

15. If the refractive index of water is $\frac{4}{3}$ and that of glass slab is $\frac{5}{3}$, then the critical angle of incidence for which a light ray tending to go from glass to water is totally reflected is :-

(1) $\sin^{-1}\left(\frac{3}{4}\right)$ (2) $\sin^{-1}\left(\frac{3}{5}\right)$
(3) $\sin^{-1}\left(\frac{2}{3}\right)$ (4) $\sin^{-1}\left(\frac{4}{5}\right)$

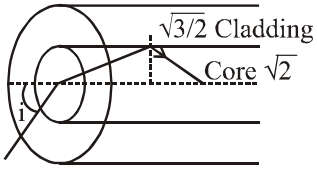
PRI235

16. With respect to air, critical angle in a medium for light of red colour [λ_1] is θ. Other facts remaining same, critical angle for light of yellow colour [λ_2] will be :-

(1) θ (2) More than θ
(3) Less than θ (4) $\frac{\theta\lambda_1}{\lambda_2}$

PRI236

17. Figure shows a cross section of a 'light pipe' made of a glass fibre of refractive index $\sqrt{2}$. The outer covering of the pipe is made of a material of refractive index $\sqrt{\frac{3}{2}}$. What is maximum angle of the incident ray with the axis of the pipe for which total internal reflections inside the pipe take place:-



(1) 45° (2) 66° (3) 80° (4) 90°

PRI237

18. A converging lens of focal length 20 cm and made of a material with refractive index 1.1 is immersed in water. The lens will behave as a:
(1) Convergent lens of focal length 20 cm
(2) Convergent lens of focal length lesser than 20 cm
(3) Divergent lens
(4) Plane glass plate

PRI238

19. An object is located at 10 cm in front of a convex lens of focal length 12 cm. Find the position and nature of image: -
(1) 60 cm real (2) 60 cm virtual
(3) 5.45 cm virtual (4) 5.45 cm real

PRI239

20. A lamp is placed 6.0 m from a wall. On putting a lens between the lamp and the wall, at a distance of 1.2 m from the lamp, a real image of the lamp is formed on the wall. The magnification of the image is :-
(1) 3 (2) 4 (3) 5 (4) 6

PRI240

21. A plano convex and another plano convex lens have the same focal length of 20 cm and are put in contact to form a lens combination. The combination is used to view an object of 0.5 cm length kept at 20 cm form the lens combination. As compared to the object, the image will be :-
(1) Magnified and inverted
(2) Reduced and erect
(3) Same size as the object and erect
(4) Same size as the object and inverted

PRI241

22. A concave mirror and a convex lens both have same focal length in air. When they are submerged in glycerine, the focal length :-
(1) Of both still remains unchanged
(2) Of the mirror remains unchanged while that of the lens increases
(3) Of the mirror remains unchanged while that of the lens decreases
(4) Of both may increase or decrease

PRI242

23. Two plano-concave lenses of glass of refractive index 1.5 have radii of curvature of 20 and 30 cm. They are placed in contact with curved surfaces towards each other and the space between them is filled with a liquid of refractive index (4/3). Find the focal length of the system.
(1) 60 cm (2) 72 cm
(3) 40 cm (4) 34 cm

PRI243

24. An object of size 3 cm is placed 14 cm in front of a concave lens of focal length 21 cm. Describe the image produced by the lens. **[NCERT]**
(1) Virtual, erect, diminished
(2) Virtual, erect, enlarged
(3) Real, inverted, enlarged
(4) Real, inverted, diminished

PRI244

25. What is the focal length when a convex lens of focal length 30 cm in contact with a concave lens of focal length 20 cm ? **[NCERT]**
(1) –12 cm (2) 12 cm
(3) –60 cm (4) +60 cm

PRI245

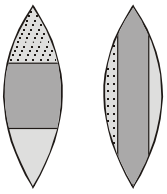
26. Determine the nature of lens shown in figure (P) & (Q) (given lenses are made from glass & surrounding medium is air)



(1) P → converging & Q → diverging
(2) P → diverging & Q → converging
(3) P & Q both converging
(4) P & Q both diverging

PRI246

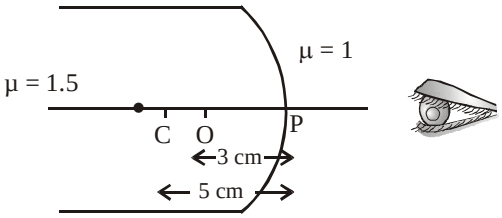
27. Figure shows two convex lenses A and B, each made up of three different transparent materials. The number of images formed, of an object kept on the principal axis of each lens will be



(1) 3 and 3 (2) 3 and 1
(3) 1 and 1 (4) 3 and 2

PRI247

28. A bubble in glass (μ = 1.5) is situated at a distance 3 cm from a convex surface of diameter 10 cm as shown in figure At what distance from the surface will the bubble appear:



(1) 2.5 cm (2) 2 cm
(3) 3 cm (4) 4 cm

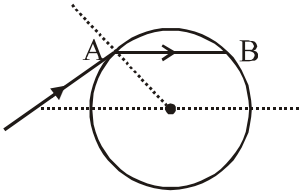
PRI248

29. One end of a cylindrical glass rod (μ = 1.5) is in the shape of a concave refraction surface of radius 10 cm. An air bubble is situated in the glass rod at a point on its axis such that it appears to be at a distance 10 cm from the surface inside the glass when seen from the other medium. find the actual location of air bubble from the surface.

(1) 10 cm inside the glass
(2) 20 cm outside the glass
(3) 30 cm inside the glass
(4) 40 cm inside the glass

PRI249

30. What should be the angle of incidence at A of the spherical glass placed in air so that grazing emergence of light ray takes place at B?



(1) 30° (2) 45° (3) 60° (4) 90°

PRI250

31. A ray of light is incident at an angle of 60° on one face of a prism of angle 30°. The emergent ray of light makes an angle of 30° with incident ray. The angle made by emergent ray with second face of prism will be: -

(1) 0° (2) 90° (3) 45° (4) 30°

PRI251

32. When a ray of light is incident at angle of 35° on an equilateral prism. Then it passes symmetrically. Calculate the deviation of the ray. (Symmetrically means parallel to base)

(1) 30° (2) 15°
(3) 10° (4) 45°

PRI252

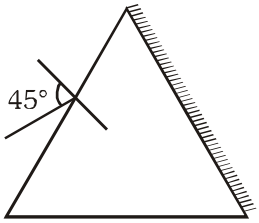
33. A prism is made of glass of unknown refractive index. A parallel beam of light is incident on a face of the prism. The angle of minimum deviation is measured to be 30°. What is refractive index of the material of the prism?

(Refracting angle of prism is 60°)

(1) 1.414 (2) 1.5
(3) 1.7 (4) None of these

PRI253

34. One face of a prism with a refracting angle 30° is coated with silver. A ray is incident on another face at an angle of 45° is refracted and reflected from the silver coated face & retraces its path. The refractive index of prism is



(1) $\sqrt{2}$ (2) 2 (3) $\sqrt{3}$ (4) 1.5

PRI254

35. A man uses a convex lens of focal length 5 cm as magnifying glass. The least distance of distinct vision of the man is 25 cm. Calculate the magnification if the image is formed at the least distance of distinct vision.

(1) 1.5 (2) 3 (3) 4 (4) 6

PRI255

36. An astronomical telescope has an eye piece of focal length 5 cm. If magnification produced is 14 in normal adjustment, then calculate the length of the telescope.

(1) 75cm (2) 9cm (3) 50cm (4) 55cm

PRI256

37. A compound microscope consists of an objective lens of focal length 2 cm and an eye piece of focal length 6.25 cm separated by a distance of 15 cm. How far from the objective should an object be placed in order to obtain the final image at the least distance of distinct vision (25 cm). **[NCERT]**

(1) 2.5 cm (2) 1.67 cm

(3) 2 cm (4) 3.3 cm

PRI257

38. A person with a normal near point (25 cm) using a compound microscope with objective of focal length 8.0 mm and an eye piece of focal length 2.5 cm can bring an object placed at 9 mm from the objective in sharp focus. What is the separation between two lenses?

[NCERT]

(1) 9.46 cm (2) 4.85 cm

(3) 5.73 cm (4) None of these

PRI258

39. A giant refracting telescope at an observatory has an objective lens of focal length 15 m. If an eye piece of focal length 1.0 cm is used What is the angular magnification of the telescope: -

[NCERT]

(1) 1.5 (2) 1500 (3) 15 (4) 150

PRI259

40. A man with normal near point (25 cm) reads a book with small print using a magnifying glass : a thin convex lens of focal length 5 cm. What is the closest and the farthest distance at which he should keep the lens from the page so that he can read the book when viewing through the magnifying glass?

[NCERT]

(1) –4.8 cm, –5.2 cm (2) –4.2 cm, –5 cm

(3) –6.2 cm, –8 cm (4) –9 cm, –2.9 cm

PRI260

41. A card sheet divided into square each of size 1 mm^2 is being viewed at a distance of 9 cm through a magnifying glass (a converging lens of focal length 10 cm) held close to the eye. What is the magnification produced by the lens?

[NCERT]

(1) $\frac{10}{19}$ (2) 9 (3) 10 (4) 5

PRI261

42. An angular magnification (magnifying power) of 30X is desired using an objective of focal length 1.25 cm and an eye piece of focal length 5 cm. How will you setup the compound microscope for normal adjustment?

[NCERT]

(1) Separation between both lenses 13.75 cm

(2) Separation between both lenses 15.51 cm

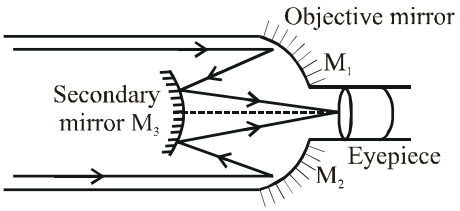
(3) Separation between both lenses 6.25 cm

(4) Separation between both lenses 9.0 cm

PRI262

43. An instrument uses two mirrors as shown in figure. Such instrument is built with the mirrors 20 mm apart. If the radius of curvature of the large mirror is 220 mm and the small mirror is 140 mm. Where will the final image from M_3 of an object at infinity be?

[NCERT]



(1) 21 cm (2) 31.5 cm

(3) 27.5 cm (4) 19 cm

PRI263

44. Two identical thin biconvex lenses of focal length 15 cm and refractive index 1.5 are in contact with each other. The space between the lenses is filled with a liquid of refractive index 1.25. The focal length of the combination is _____ cm.

PRI264

45. The difference of speed of light in the two media A and B ($v_A - v_B$) is 2.6×10^7 m/s. If the refractive index of medium B is 1.47, then the ratio of refractive index of medium B to medium A is : (Given : speed of light in vacuum $c = 3 \times 10^8$ ms^{-1})

(1) 1.303 (2) 1.318

(3) 1.13 (4) 0.12

PRI265

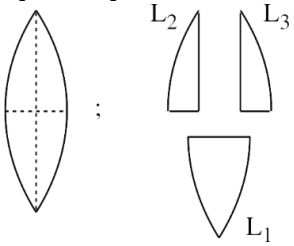
46. Consider a light ray travelling in air is incident into a medium of refractive index $\sqrt{2n}$. The incident angle is twice that of refracting angle. Then, the angle of incidence will be :

(1) $\sin^{-1}\left(\sqrt{n}\right)$ (2) $\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

(3) $\sin^{-1}\left(\sqrt{2n}\right)$ (4) $2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

PRI266

47. A convex lens has power P. It is cut into two halves along its principal axis. Further one piece (out of the two halves) is cut into two halves perpendicular to the principal axis (as shown in figure). Choose the incorrect option for the reported pieces.



(1) Power of $L_1 = \frac{P}{2}$

(2) Power of $L_2 = \frac{P}{2}$

(3) Power of $L_3 = \frac{P}{2}$

(4) Power of $L_1 = P$

PRI267

48. The refracting angle of a prism is A and refractive index of the material of the prism is cot(A/2). Then the angle of minimum deviation will be -

(1) 180 – 2A (2) 90 – A

(3) 180 + 2A (4) 180 – 3A

PRI268

49. The speed of light in media 'A' and 'B' are 2.0×10^{10} cm/s and 1.5×10^{10} cm/s respectively. A ray of light enters from the medium B to A at an incident angle 'θ'. If the ray suffers total internal reflection, then

(1) $\theta = \sin^{-1}\left(\frac{3}{4}\right)$ (2) $\theta > \sin^{-1}\left(\frac{2}{3}\right)$

(3) $\theta < \sin^{-1}\left(\frac{3}{4}\right)$ (4) $\theta > \sin^{-1}\left(\frac{3}{4}\right)$

PRI269

50. A convex lens of focal length 20 cm is placed in front of convex mirror with principal axis coinciding each other. The distance between the lens and mirror is 10 cm. A point object is placed on principal axis at a distance of 60 cm from the convex lens. The image formed by combination coincides the object itself. The focal length of the convex mirror is ____ cm.

PRI270

51. Light travels in two media M_1 and M_2 with speeds 1.5×10^8 ms^{-1} and 2.0×10^8 ms^{-1} respectively. The critical angle between them is:

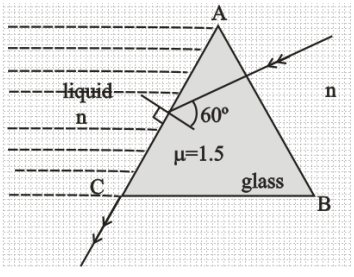
(1) $\tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$ (2) $\tan^{-1}\left(\frac{2}{3}\right)$

(3) $\cos^{-1}\left(\frac{3}{4}\right)$ (4) $\sin^{-1}\left(\frac{2}{3}\right)$

PRI271

52. In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index 'n'. For incident angle 60° at the side AC, the refracted light beam just grazes along face AC. The refractive index of the liquid $n = \frac{\sqrt{x}}{4}$. The value of x is_____.

(Given refractive index of glass = 1.5)



PRI272

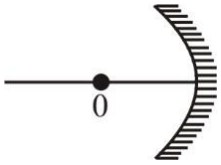
53. In normal adjustment, for a refracting telescope, the distance between objective and eye piece is 30 cm. The focal length of the objective, when the angular magnification of the telescope is 2, will be:

(1) 20 cm (2) 30 cm

(3) 10 cm (4) 15 cm

PRI273

54. An object 'o' is placed at a distance of 100 cm in front of a concave mirror of radius of curvature 200 cm as shown in the figure. The object starts moving towards the mirror at a speed 2 cm/s. The position of the image from the mirror after 10s will be at ..... cm.



PRI274

55. Light enters from air into a given medium at an angle of 45° with interface of the air-medium surface. After refraction, the light ray is deviated through an angle of 15° from its original direction. The refractive index of the medium is :

(1) 1.732 (2) 1.333

(3) 1.414 (4) 2.732

PRI275

56. A convex lens of refractive index 1.5 and focal length 18 cm in air is immersed in water. The change in focal length of the lens will be____cm.

(Given refractive index of water = $\frac{4}{3}$)

PRI276

57. The light rays from an object have been reflected towards an observer from a standard flat mirror, the image observed by the observer is :-

A. Real

B. Erect

C. Smaller in size then object

D. Laterally inverted

Choose the most appropriate answer from the options given below :

(1) B and D only (2) B and C only

(3) A and D only (4) A, C and D only

PRI277

58. A microscope is focused on an object at the bottom of a bucket. If liquid with refractive index $\frac{5}{3}$ is poured inside the bucket, then microscope have to be raised by 30 cm to focus the object again. The height of the liquid in the bucket is :

(1) 75 cm (2) 50 cm

(3) 18 cm (4) 12 cm

PRI278

ANSWER KEY

1.	(2)	**2.**	(2)	**3.**	(3)	**4.**	(1)	**5.**	(3)
6.	(1)	**7.**	(1)	**8.**	(1)	**9.**	(2)	**10.**	(3)
11.	(1)	**12.**	(3)	**13.**	(1)	**14.**	(4)	**15.**	(4)
16.	(3)	**17.**	(1)	**18.**	(3)	**19.**	(2)	**20.**	(2)
21.	(4)	**22.**	(2)	**23.**	(2)	**24.**	(1)	**25.**	(3)
26.	(3)	**27.**	(2)	**28.**	(1)	**29.**	(3)	**30.**	(4)
31.	(2)	**32.**	(3)	**33.**	(1)	**34.**	(1)	**35.**	(4)
36.	(1)	**37.**	(1)	**38.**	(1)	**39.**	(2)	**40.**	(2)
41.	(3)	**42.**	(1)	**43.**	(2)	**44.**	10	**45.**	(3)
46.	(4)	**47.**	(1)	**48.**	(1)	**49.**	(4)	**50.**	10
51.	(1)	**52.**	27	**53.**	(1)	**54.**	400	**55.**	(3)
56.	54	**57.**	(1)	**58.**	(1)				

Wave Optics

1. In an interference pattern, at a point we observe the 16th order bright fringe or maxima for λ_1 = 7200Å. What order will be visible here if the source is replaced by light of wavelength λ_2 = 4800Å.

(1) 12 (2) 24

(3) 36 (4) 16

PWO197

2. In the Young's double slit experiment, the interference pattern is found to have an intensity ratio between bright and dark fringes, as 9. This implies that :-

(1) the intensities ratio of sources is 5

(2) the intensities ratio of sources is 4

(3) the amplitude ratio of sources is 3

(4) the amplitude ratio of sources is 2

PWO198

3. White light is used to illuminate the two slits in a Young's double slit experiment. The separation between slits is b and the screen is at a distance d (>>b) from the slits. At a point on the screen directly infront of one of the slits, certain wavelengths are missing. Some of these missing wavelengths are :-

(1) $\lambda = b^2/d$ (2) $\lambda = 2b^2/d$

(3) $\lambda = b^2/3d$ (4) $\lambda = 2b^2/3d$

PWO199

4. Two light rays having the same wavelength λ in vacuum are in phase initially. Then the first ray travels a path of length L_1 through a medium of refractive index n_1 while the second ray travels a path of length L_2 through a medium of refractive index n_2. The two waves are then combined to observe interference. The phase-difference between the two waves is :-

(1) $\frac{2\pi}{\lambda}(L_2 - L_1)$ (2) $\frac{2\pi}{\lambda}(n_1L_1 - n_2L_2)$

(3) $\frac{2\pi}{\lambda}(n_2L_1 - n_1L_2)$ (4) $\frac{2\pi}{\lambda}\left(\frac{L_1}{n_1} - \frac{L_2}{n_2}\right)$

PWO200

5. In two separate setup of the Young's double slit experiment, fringes of equal width are observed when lights of wavelengths in the ratio 1 : 2 are used. If the ratio of the slit separation in the two cases is 2 : 1, the ratio of the distances between the plane of the slits and the screen, in the two setup is :-

(1) 4 : 1 (2) 1 : 1 (3) 1 : 4 (4) 2 : 1

PWO201

6. A thin sheet of glass (μ = 1.5) of thickness 6 microns introduced in the path of one of interfering beams in a double slit experiment shifts the central fringe to a position previously occupied by fifth bright fringe. Then the wavelength of light used is :-

(1) 6000 Å (2) 3000 Å

(3) 4500 Å (4) 7500 Å

PWO202

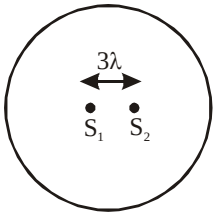
7. The central fringe of the interference pattern produced by light of wavelength 6000 Å is found to shift to the position of 4th bright fringe after a glass plate of refractive index 1.5 is introduced in front of one slit in Young's experiment. The thickness of the glass plate will be :-

(1) 4.8 µm (2) 8.23 µm

(3) 14.98 µm (4) 3.78 µm

PWO203

8. If two coherent sources are placed at a distance 3λ from each other symmetric to the centre of the circle shown in the figure, then number of maxima shown on the screen placed along the circumference is :-



(1) 16 (2) 12

(3) 8 (4) 4

PWO204

9. Light of wavelength λ is incident on a slit of width d. The resulting diffraction pattern is observed on a screen at a distance D. The linear width of the central maxima equal to width of the slit if D equals

(1) d/λ (2) $2\lambda/d$

(3) $d^2/2\lambda$ (4) $2\lambda^2/d$

PWO205

10. The main difference in the phenomenon of interference and diffraction is that :-

(1) diffraction is due to interaction of light from the same wavefront whereas interference is the interaction of waves from two isolated sources.

(2) diffraction is due to interaction of light from wavefront, whereas the interference is the interaction of two waves derived from the same source.

(3) diffraction is due to interaction of waves derived from the same source, whereas the interference is the bending of light from the same wavelength.

(4) diffraction is caused by reflected waves from a source wheras interference caused is due to refraction of waves from a surface.

PWO206

11. A ray of light strikes a glass plate at an angle of 60°. If the reflected and refracted rays are perpendicular to each other, the index of refraction of glass is :-

(1) $\sqrt{3}$ (2) 3/2 (3) $\sqrt{3/2}$ (4) 1/2

PWO207

12. When unpolarised light is incident on a plane glass plate at Brewster's (polarising) angle, then which of the following statements is correct :-

(1) reflected and refracted rays are completely polarised with their planes of polarisation parallel to each other.

(2) reflected and refracted rays are completely polarised with their planes of polarisation perpendicular to each other

(3) the reflected light is plane polarised

(4) the reflected light is partially polarised but refracted light is plane polarised

PWO208

13. In a Fraunhoffer diffraction experiment at a single slit using light of wavelength 400 nm, the first minimum is formed at an angle of 30°. Then the direction θ of the first secondary maximum is :-

(1) $\tan^{-1}\left(\frac{4}{3}\right)$ (2) 60°

(3) $\sin^{-1}\left(\frac{3}{4}\right)$ (4) $\tan^{-1}\left(\frac{3}{4}\right)$

PWO209

14. Two light beams of intensities in the ratio of 9 : 4 are allowed to interfere. The ratio of the intensity of maxima and minima will be :

(1) 2 : 3 (2) 16 : 81

(3) 25 : 169 (4) 25 : 1

PWO210

15. The two light beams having intensities I and 9I interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\frac{\pi}{2}$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be :

(1) 2 I (2) 6 I (3) 5 I (4) 7 I

PWO211

16. The interference pattern is obtained with two coherent light sources of intensity ratio 4 :1. And the ratio $\frac{I_{max}+I_{min}}{I_{max}-I_{min}}$ is $\frac{5}{x}$. Then, the value of x will be equal to :

(1) 3 (2) 4 (3) 2 (4) 1

PWO212

17. In Young's double slit experiment the two slits are 0.6 mm distance apart. Interference pattern is observed on a screen at a distance 80 cm from the slits. The first dark fringe is observed on the screen directly opposite to one of the slits. The wavelength of light will be ____ nm.

PWO213

18. Using Young's double slit experiment, a monochromatic light of wavelength 5000 Å produces fringes of fringe width 0.5 mm. If another monochromatic light of wavelength 6000Å is used and the separation between the slits is doubled, then the new fringe width will be :

(1) 0.5 mm (2) 1.0 mm

(3) 0.6 mm (4) 0.3 mm

PWO214

19. In young's double slit experiment, the fringe width is 12mm. If the entire arrangement is placed in water of refractive index $\frac{4}{3}$, then the fringe width becomes (in mm)

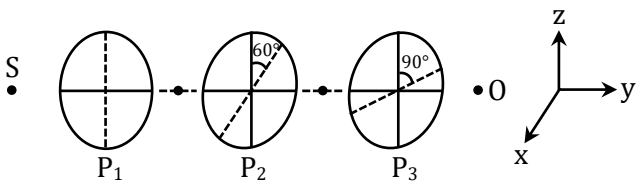
(1) 16 (2) 9 (3) 48 (4) 12

PWO215

20. Two light beams of intensities 4I and 9I interfere on a screen. The phase difference between these beams on the screen at point A is zero and at point B is π. The difference of resultant intensities, at the point A and B, will be _____ I.

PWO216

21. As shown in figures, three identical polaroids P_1, P_2 and P_3 are placed one after another. The pass axis of P_2 and P_3 are inclined at angle of 60° and 90° with respect to axis of P_1. The source S has an intensity of $256\frac{W}{m^2}$. The intensity of light at point O is $-----\frac{W}{m^2}$.



PWO217

22. Two light waves of wavelengths 800 and 600 nm are used in Young's double slit experiment to obtain interference fringes on a screen placed 7 m away from plane of slits. If the two slits are separated by 0.35 mm, then shortest distance from the central bright maximum to the point where the bright fringes of the two wavelength coincide will be _____mm.

PWO218

ANSWER KEY

1.	(2)	**2.**	(2,4)	**3.**	(1,3)	**4.**	(2)	**5.**	(1)
6.	(1)	**7.**	(1)	**8.**	(2)	**9.**	(3)	**10.**	(2)
11.	(1)	**12.**	(3)	**13.**	(3)	**14.**	(4)	**15.**	(2)
16.	(2)	**17.**	450	**18.**	(4)	**19.**	(2)	**20.**	24
21.	24	**22.**	48						

Modern Physics

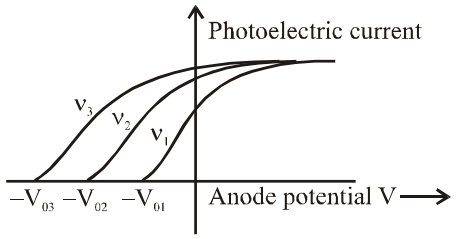
1. Consider a parallel beam of light of wavelength 600 nm and intensity 100 W/m^2. How many photons cross 1cm^2 area perpendicular to beam in one second?
(1) 3×10^{16} (2) 3×10^{14}
(3) 3×10^{9} (4) 3×10^{4}

PMD256

2. When light of wavelength 300 nm falls on a photoelectric emitter, photoelectrons are just liberated. For another emitter, light of wavelength 600 nm is just sufficient for liberating photoelectrons. The ratio of the work function of the two emitters is :-
(1) 1 : 2 (2) 2 : 1 (3) 4 : 1 (4) 1 : 4

PMD257

3. In photoelectric effect, the curve between photoelectric current and anode potential V (For different frequencies) is shown in figure, then



(1) $\nu_1 > \nu_2 > \nu_3$ (2) $\nu_1 < \nu_2 < \nu_3$
(3) $\nu_1 = \nu_2 = \nu_3$ (4) $\nu_1 > \nu_2 < \nu_3$

PMD258

4. Wave nature of light cannot explain photoelectric effect because in photoelectric effect, it is seen that :
(1) For the frequency of light below a certain value, the photoelectric effect does not take place, irrespective of intensity
(2) Maximum kinetic energy of ejected electrons is independent of intensity of radiation
(3) There is no time lag between the incidence of radiation and emission of electrons
(4) All of these

PMD259

5. A monochromatic point source of light is placed at a distance d from a metal surface. Photoelectrons are ejected at a rate n per second, and with maximum kinetic energy E. If the source is brought nearer to distance d/2, the rate and the maximum kinetic energy per photoelectron become nearly
(1) 2n and 2E (2) 4n and 4E
(3) 4n and E (4) n and 4E

PMD260

6. Maximum velocity of the photoelectrons emitted by a metal surface is 1.2×10^6 ms^{-1}. Assuming the specific charge of the electron to be 1.8×10^{11} C kg^{-1}, the value of stopping potential (in volt) will be :-
(1) 2 (2) 3 (3) 4 (4) 6

PMD261

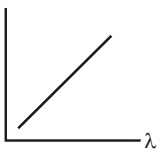
7. Given that a photon of light of wavelength 10000Å has energy 1.23 eV. Now when light of intensity I_0 and wavelength 5000 Å falls on a photo cell the saturation current and stopping potential are 0.40 μA and 1.36 V respectively. The work function is:-
(1) 0.43 eV (2) 1.10 eV
(3) 1.36 eV (4) 2.47 eV

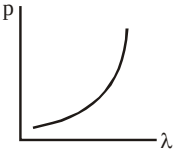
PMD262

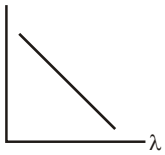
8. Two identical metal plates show photoelectric effect by a light of wavelength λ_A falling on plate A and λ_B on plate B ($\lambda_A = 2\lambda_B$). The maximum kinetic energy is :-
(1) $2K_A = K_B$ (2) $K_A < K_B/2$
(3) $K_A = 2K_B$ (4) $K_A > K_B/2$

PMD263

9. Which of the following figures represents the variation of particle momentum (p) and associated de Broglie wavelength (λ) ?

(1) 

(2) 

(3) 

(4) 

PMD264

10. What will be the ratio of de-Broglie wavelength of proton and α-particle of same energy :
(1) 2 : 1 (2) 1 : 2 (3) 4 : 1 (4) 1 : 4

PMD265

11. The de-Broglie wavelength of photoelectrons is 1Å. Its accelerating potential is :
(1) 150 V (2) 15.3 V
(3) 12.3 V (4) 13.6 V

PMD266

12. De-broglie wavelength of a body of mass 1 kg moving with velocity of 2000 m/s is :
(1) 3.32×10^{-27} Å (2) 1.5×10^{7} Å
(3) 0.55×10^{-22} Å (4) 2×10^{20}Å

PMD267

13. Find an expression for de Broglie wavelength for an electron in the nth orbit of the Bohr model of the hydrogen atom and hence find its value when the electron is in n = 4 level.
Take radius of nth orbital of Bohr's model of H-atom; $r_n = (53\ n^2)$ pm
(1) 1.33 nm (2) 1.20 nm
(3) 1.50 nm (4) None

PMD268

14. What are the respective number α- and β-particles emitted in the following radioactive decay?

$$^{200}_{90}X \rightarrow {}^{168}_{80}Y$$

(1) 6 and 8 (2) 6 and 6
(3) 8 and 8 (4) 8 and 6

PMD269

15. In the equation : $^{27}_{13}Al + {}^{4}_{2}He \longrightarrow {}^{30}_{15}P + X$,
The correct symbol for X is :-
(1) $^{0}_{-1}e$ (2) $^{1}_{1}H$ (3) $^{4}_{2}He$ (4) $^{1}_{0}n$

PMD270

16. A nucleus at rest breaks into two nuclear parts which have their velocities ratio equal to 2 : 1. What will be the ratio of their nuclear radii ?
(1) $2^{1/3} : 1$ (2) $1 : 2^{1/3}$
(3) $2^{3/2} : 1$ (4) $1 : 2^{3/2}$

PMD271

17. The binding energies of the nuclei A and B are E_a and E_b respectively. Three atoms of the element B fuse to give one atom of element A and an energy Q is released. Then E_a, E_b and Q are related :-
(1) $E_a - 3E_b = Q$ (2) $3E_b - E_a = Q$
(3) $E_a + 3E_b = Q$ (4) $E_b + 3E_a = Q$

PMD272

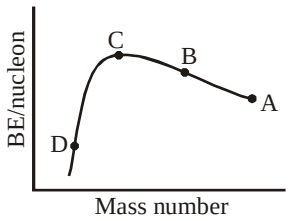
18. In a fission reaction $^{236}_{90}U \rightarrow {}^{117}X + {}^{117}Y + n + n$
The binding energy per nucleon of X and Y are 8.5 MeV whereas of ^{236}U is 7.6 MeV. The total energy liberated will be about :
(1) 400 MeV (2) 200 MeV
(3) 300 MeV (4) 200 keV

PMD273

19. The atomic mass of $_7N^{15}$ is 15.000108 a.m.u. and that of $_8O^{16}$ is 15.994915 a.m.u. If the mass of a proton is 1.007825 a.m.u. then the minimum energy provided to remove the least tightly bound proton is :-
(1) 0.013018 MeV (2) 12.13 MeV
(3) 13.018 MeV (4) 12.13 eV

PMD274

20. Binding energy per nucleon versus mass number curve for nuclei is shown in the figure. A, B, C and D are four nuclei indicated on the curve.
The process that would release energy is :



(1) C → 2D (2) A → C + D
(3) A → B + C (4) B → C + D

PMD275

21. Highly energetic electrons are bombarded on a target of an element containing 30 neutrons. The ratio of radii of nucleus to that of Helium nucleus is $14^{1/3}$. The atomic number of nucleus will be:-
(1) 25 (2) 26 (3) 56 (4) 30

PMD276

22. When light of frequency twice the threshold frequency is incident on the metal plate, the maximum velocity of emitted election is v_1. When the frequency of incident radiation is increased to five times the threshold value, the maximum velocity of emitted electron becomes v_2. If $v_2 = x\ v_1$, the value of x will be _____.

PMD277

23. Given below are two statements :-

Statement I : Davisson-Germer experiment establishes the wave nature of electrons.

Statement II : If electrons have wave nature, they can interfere and show diffraction.

In the light of the above statements choose the **correct answer** from the options given below:-

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

PMD278

24. A proton, a neutron, an electron and an α-particle have same energy. If $\lambda_p, \lambda_n, \lambda_e$ and λ_α are the de Broglie's wavelengths of proton, neutron, electron and α particle respectively, then choose the correct relation from the following :

(1) $\lambda_p = \lambda_n > \lambda_e > \lambda_\alpha$ (2) $\lambda_\alpha < \lambda_n < \lambda_p < \lambda_e$

(3) $\lambda_e < \lambda_p = \lambda_n > \lambda_\alpha$ (4) $\lambda_e = \lambda_p = \lambda_n = \lambda_\alpha$

PMD279

25. How many alpha and beta particles are emitted when Uranium $_{92}U^{238}$ decays to lead $_{82}Pb^{206}$?

(1) 3 alpha particles and 5 beta particles

(2) 6 alpha particles and 4 beta particles

(3) 4 alpha particles and 5 beta particles

(4) 8 alpha particles and 6 beta particles

PMD280

26. An α particle and a carbon 12 atom has same kinetic energy K. The ratio of their de-Broglie wavelength $(\lambda_a : \lambda_{C12})$ is :

(1) $1:\sqrt{3}$ (2) $\sqrt{3}:1$

(3) 3 : 1 (4) $2:\sqrt{3}$

PMD281

27. A beam of monochromatic light is used to excite the electron in Li^{++} from the first orbit to the third orbit. The wavelength of monochromatic light is found to be $x \times 10^{-10}$ m. The value of x is _____ .

[Given hc = 1242 eV nm]

PMD282

28. The de Broglie wavelengths for an electron and a photon are λ_e and λ_p respectively. For the same kinetic energy of electron and photon. which of the following presents the correct relation between the de Broglie wavelengths of two?

(1) $\lambda_p \propto \lambda_e^2$ (2) $\lambda_p \propto \lambda_e$

(3) $\lambda_p \propto \sqrt{\lambda_e}$ (4) $\lambda_p \propto \sqrt{\frac{1}{\lambda_e}}$

PMD283

29. The electric field at the point associated with a light wave is given by

$E = 200 [\sin(6 \times 10^{15}) t + \sin(9 \times 10^{15}) t]$ Vm^{-1}

Given : $h = 4.14 \times 10^{-15}$ eVs

If this light falls on a metal surface having a work function of 2.50 eV, the maximum kinetic energy of the photoelectrons will be :

(1) 1.90 eV (2) 3.27 eV

(3) 3.60 eV (4) 3.42 eV

PMD284

30. The ratio of wavelengths of proton and deuteron accelerated by potential V_p and V_d is $1:\sqrt{2}$. Then, the ratio of V_p to V_d will be

(1) 1 : 1 (2) $\sqrt{2}:1$ (3) 2 : 1 (4) 4 : 1

PMD285

31. Mass numbers of two nuclei are in the ratio of 4:3. Their nuclear densities will be in the ratio of

(1) 4:3 (2) $\left(\frac{3}{4}\right)^{\frac{1}{3}}$

(3) 1 : 1 (4) $\left(\frac{4}{3}\right)^{\frac{1}{3}}$

PMD286

32. An electron (mass m) with an initial velocity $\vec{v} = v_0\hat{i}(v_0 > 0)$ is moving in an electric field $\vec{E} = -E_0\hat{i}(E_0 > 0)$ where E_0 is constant. If at t = 0 de Broglie wavelength is $\lambda_0 = \frac{h}{mv_0}$, then its de Broglie wavelength after time t is given by

(1) λ_0

(2) $\lambda_0\left(1+\frac{eE_0t}{mv_0}\right)$

(3) $\lambda_0 t$

(4) $\frac{\lambda_0}{\left(1+\frac{eE_0t}{mv_0}\right)}$

PMD287

33. The ratio of de-Broglie wavelength of an α-particle and a proton accelerated from rest by the same potential is $\frac{1}{\sqrt{m}}$, the value of m is

(1) 4

(2) 16

(3) 8

(4) 2

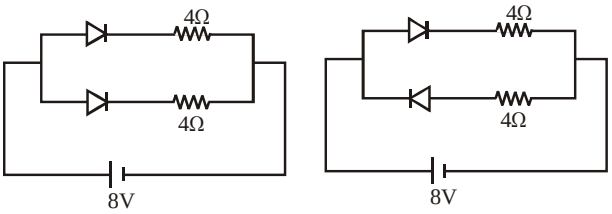
PMD288

ANSWER KEY

1.	(1)	**2.**	(2)	**3.**	(2)	**4.**	(4)	**5.**	(3)
6.	(3)	**7.**	(2)	**8.**	(2)	**9.**	(4)	**10.**	(1)
11.	(1)	**12.**	(1)	**13.**	(1)	**14.**	(4)	**15.**	(4)
16.	(2)	**17.**	(1)	**18.**	(2)	**19.**	(2)	**20.**	(3)
21.	(2)	**22.**	2	**23.**	(1)	**24.**	(2)	**25.**	(4)
26.	(2)	**27.**	114	**28.**	(1)	**29.**	(4)	**30.**	(4)
31.	(3)	**32.**	(4)	**33.**	(3)				

Semiconductor Electronics

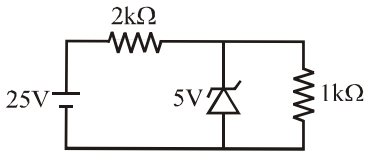
1. Current following in each of the circuit A & B respectively are: -



(1) 1 A, 2 A (2) 1 A, 4 A
(3) 4 A, 2 A (4) 2 A, 4 A

PSE206

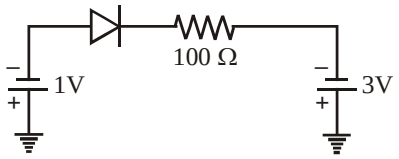
2. Find current through zener diode :-



(1) 5mA (2) 7.5 mA
(3) 2.5 mA (4) 12.5 mA

PSE207

3. What is the current through an ideal p-n junction diode shown in figure below: -



(1) Zero (2) 10 mA
(3) 20 mA (4) 50 mA

PSE208

4. Potential drop across a forward biased p-n junction diode is 0.7 V. If a battery of 4 V is applied, calculate the minimum resistance to be put in series, for the maximum current of 5 mA in the circuit.

(1) 660 Ω (2) 350 Ω
(3) 475 Ω (4) 500 Ω

PSE209

5. A light emitting diode (LED) has a voltage drop of 2 V across it and passes a current of 10 microampere, when it is operated with a 6 V battery through a limiting resistor R. The value of R is :

(1) 10 kΩ (2) 4 kΩ
(3) 200 kΩ (4) 400 kΩ

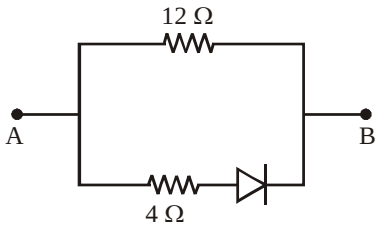
PSE210

6. In an intrinsic semiconductor, the density of conduction electrons is 7.07×10^{15} m^{-3}. When it is doped with indium, the density of holes becomes 5×10^{22} m^{-3}. Find the density of conduction electrons in doped semiconductor :

(1) Zero (2) 1×10^{9} m^{-3}
(3) 7×10^{15} m^{-3} (4) 5×10^{22} m^{-3}

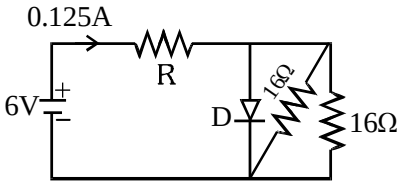
PSE211

7. Find the equivalent resistance of the network shown in the figure between the points A and B, if (i) $V_A > V_B$ (ii) $V_A < V_B$



PSE212

8. In the given circuit, the diode has a forward resistance of 8 Ω and infinite backward resistance. The value of R is :-



(1) 44 Ω (2) 48 Ω (3) 16 Ω (4) 8 Ω

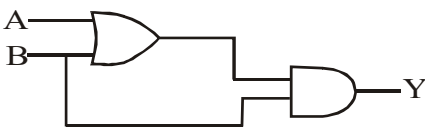
PSE213

9. Which of the following is not equal to 1 in Boolean algebra :-

(1) A + 1 (2) $A + \bar{A}$
(3) $A.\ \bar{A}$ (4) $\overline{A\ .\ \bar{A}}$

PSE214

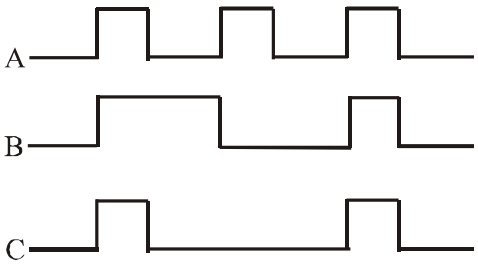
10. The output Y of the combination of gates shown is equal to :-



(1) A (2) B
(3) A + B (4) A.B

PSE215

11. For a logic gate A & B are input and C is output voltage waveform then logic gate is :-

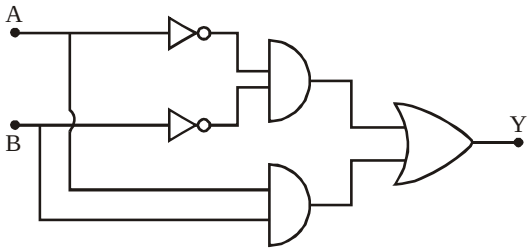


(1) OR (2) AND

(3) NOR (4) NAND

PSE216

12. Write down the boolean expression for output Y of a system shown in figure.



(1) $A\bar{B}+\bar{A}B$

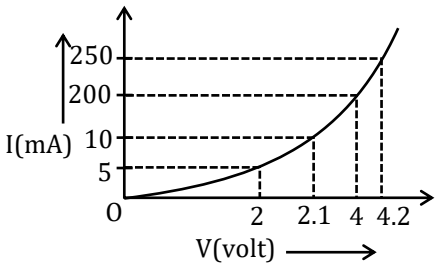
(2) $(\bar{A}-\bar{B})(A+B)$

(3) $\bar{A}\bar{B}+AB$

(4) $A\cdot B+(\bar{A}+\bar{B})$

PSE217

13. The I-V characteristics of a p-n junction diode in forward bias is shown in the figure. The ratio of dynamic resistance, corresponding to forward bias voltages of 2V and 4V respectively, is :

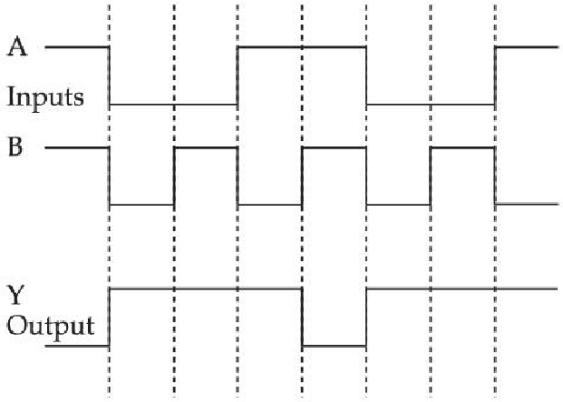


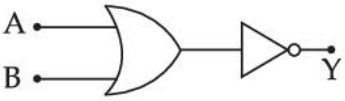
(1) 1 : 2 (2) 5 : 1

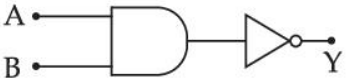
(3) 1 : 40 (4) 20 : 1

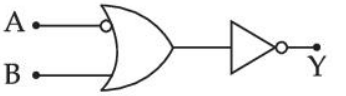
PSE218

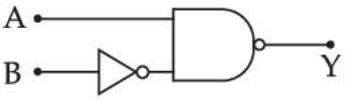
14. Identify the correct Logic Gate for the following output (Y) of two inputs A and B.



(1) 

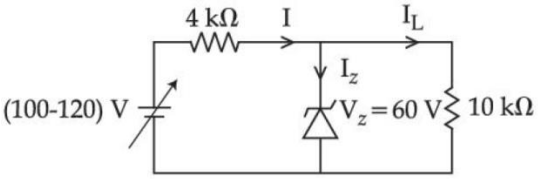
(2) 

(3) 

(4) 

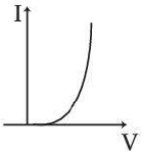
PSE219

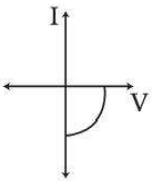
15. In the circuit shown below, maximum zener diode current will be ___mA.

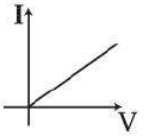


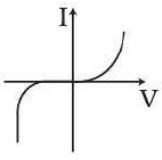
PSE220

16. Identify the solar cell characteristics from the following options :

(1) 

(2) 

(3) 

(4) 

PSE221

17. Match the List I with List II.

	List I		List II
A.	Intrinsic Semiconductor	I.	Fermi-level near conduction band
B.	n-type semiconductor	II.	Fermi-level at middle
C.	p-type semiconductor	III.	Fermi-level near valence band
D.	Metals	IV.	Fermi-level inside conduction band

Choose the **correct** answer from the options given below:

(1) (A) → I, (B) → II, (C) → III, (D) → IV

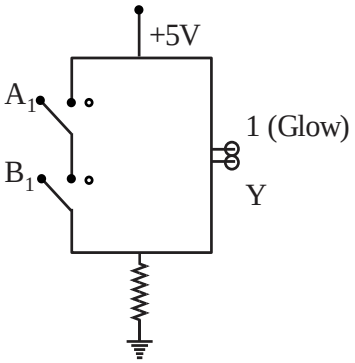
(2) (A) → II, (B) → I, (C) → III, (D) →IV

(3) (A) → II, (B) → III, (C) → I, (D) → IV

(4) (A) → III, (B) → I, (C) → II, (D) → IV

PSE222

18. The logic gate equivalent to the given circuit diagram is :



(1) OR (2) NAND

(3) NOR (4) AND

PSE223

19. **Statement I :** When a Si sample is doped with Boron, it becomes P type and when doped by Arsenic it becomes N-type semi conductor such that P-type has excess holes and N-type has excess electrons.

Statement II : When such P-type and N-type semi-conductors, are fused to make a junction, a current will automatically flow which can be detected with an externally connected ammeter.

In the light of above statements, choose the most appropriate answer from the options given below.

Options:

(1) Both Statement I and statement II are incorrect

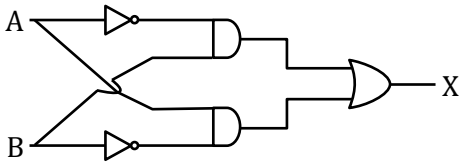
(2) Statement I is incorrect but statement II is correct

(3) Both Statement I and statement II are correct

(4) Statement I is correct but statement II is incorrect

PSE224

20. For the given logic gates combination, the correct truth table will be



(1)

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

(2)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

(3)

A	B	X
0	0	1
0	1	0
1	0	1
1	1	0

(4)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

PSE225

ANSWER KEY

1.	(3)	**2.**	(1)	**3.**	(3)	**4.**	(1)	**5.**	(4)
6.	(2)	**7.**	(i) 3Ω (ii) 12Ω			**8.**	(1)	**9.**	(3)
10.	(2)	**11.**	(2)	**12.**	(3)	**13.**	(2)	**14.**	(2)
15.	9	**16.**	(2)	**17.**	(2)	**18.**	(2)	**19.**	(4)
20.	(2)								

Important Notes